No Country for Old Men: Aging, Innovation, and Dynamic Inefficiency †

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Abstract

We study how population aging shapes the direction of innovation. In an overlapping-generations economy where older consumers face adoption costs for new technology, we show that the market over-innovates: each firm observes dampened old-age demand, but no atomistic firm internalizes how collective innovation shifts relative prices against older consumers. This technology-composition externality generates a novel dynamic inefficiency: the economy accumulates the wrong *kind* of technology, not the wrong quantity of capital. A calibrated demographic transition produces significant welfare losses borne almost entirely by the old, amplifying intergenerational inequality precisely when aging is fastest. The corrective optimal policy is a state-contingent innovation tax of 60–63% along the calibrated transition and 69–83% across plausible demographic states—far above any rate actual economies implement: the world is, empirically, no country for old men.

Keywords: Population aging, directed innovation, dynamic inefficiency, technology adoption, OLG

JEL codes: D58, D62, J11

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1 Introduction

From AI-driven services to digital banking to app-based platforms, the technological landscape increasingly reflects the preferences of younger consumers who adopt without friction. Older adults—a growing share of the population—face well-documented costs in navigating these technologies: learning unfamiliar interfaces, abandoning established routines, adapting to products not designed with them in mind. But is this simply the cost of progress, or does it reflect a market failure? We argue the latter: the market systematically over-provides new technology. Each firm observes dampened demand from older consumers and correctly anticipates demographic change, yet no atomistic firm internalizes how collective innovation shifts relative prices against older consumers who face adoption costs. The result is inefficient over-innovation.

The paper makes two contributions. First, we identify a novel demand-side channel from population aging to innovation. In an overlapping-generations economy with new-technology and retro-technology firms, age-based demand heterogeneity—young households consume new technology costlessly while old households face adoption costs—feeds back through firm profitability into innovation decisions, shaping the economy's technology composition. We call this the *technology-composition externality*. It operates through the aggregate technology mix Q_n: collective innovation shifts Q_n, which changes relative prices and pushes old consumers toward goods that are costly for them to adopt. No atomistic firm internalizes its contribution to this aggregate price shift. The deeper interpretation is that the technology choice itself carries no priced cost in the competitive equilibrium: both technologies use the same labor at the same wage, no factor market differentiates them, and no security or licensing market prices the aggregate composition Q_n. The corrective innovation tax we derive is precisely this missing price—a Pigouvian wedge that makes firms face the social marginal cost of the technology choice they are de facto making. We further show that the externality is not a generic feature of OLG-with-monopolistic-competition models: it requires (i) two technologies on a non-trivial composition margin and (ii) heterogeneous demand, while (iii) shared inputs amplify but are not strictly necessary—heterogeneous demand is the binding ingredient, and removing it eliminates the wedge regardless of factor structure (Section 5.2). Our channel complements the supply-side mechanisms emphasized in the aging-and-growth literature—labor scarcity, reduced entrepreneurship, induced automation—by showing that aging also distorts the *composition* of product demand, with aggregate welfare consequences.

Second, we show that the corrective tax required to fix this distortion is large in magnitude and that no real economy implements anything comparable on the composition margin. The externality is structurally large—60–63% along the calibrated transition and 69–83% across plausible demographic states—persistent across demographic regimes, and uncorrected on the new-vs-retro tilt anywhere in the world. The conventional remedy for OLG dynamic inefficiency—unfunded intergenerational transfers—does not address this distortion: what is needed instead is a technology policy that redirects the innovation *mix* toward retro technology, operating on a margin that is conceptually distinct from the level of R&D targeted by standard knowledge-spillover subsidies.

In a calibrated demographic transition from a younger to an older society, the over-innovation gap—the excess of new-technology firms relative to the planner's optimum—widens from 6.8% to 10.2% at the peak of the transition under the static CEP, and stays in a flatter 3%–7% range under the Markov-perfect (time-consistent) dynamic CEP. The welfare cost falls almost entirely on old households: the intergenerational welfare gap $u_Y - u_O$ spikes during the transition while young households' utility barely moves. The implied corrective policy is a state-contingent innovation tax indexed to the inherited technology stock and the demographic state—the static-CEP tax along the transition is 72%–78%, while the Markov-perfect tax is 60%–63%—ranges well above what any real country implements, leaving the world, empirically, no country for old men. A formal sensitivity analysis across symmetric demographic states delivers the paper's empirical core: the corrective tax stays in the 69–83% range as ϕ rises from 0.8 to 1.5, with the absolute externality wedge between social and private marginal cost approximately constant. The market's over-innovation is persistent, not a feature of any particular demographic regime—and no actual country implements anything close to this corrective tax. The same comparative static shows that aging depresses innovation effort by roughly a third: countries with younger demographics innovate more intensely, sustain a larger share of new-technology firms, but also face a wider over-innovation gap relative to the planner's optimum.

Related literature. *Directed technical change.* Our demand-externality mechanism builds on the theory of directed technical change (Acemoglu, 2002), in which market size and composition steer the direction of innovation. We add a lifecycle dimension: the market-size effect is dynamic, as the young's demand today shapes the technology mix they face when old. Earlier work on demand-driven innovation includes Schmookler (1966) on demand conditions and patenting, and Foellmi and Zweimüller (2006) on income distribution and product innovation. We differ in focusing on age rather than income as the source of demand heterogeneity. Our paper also connects to work on misdirected innovation: Acemoglu and Restrepo (2019) argue that firms may adopt "so-so technologies" that displace workers without sufficient productivity gains, and Acemoglu et al. (2012) make a parallel argument for clean versus dirty innovation. We share the premise that the direction of innovation matters, not just its level, but identify demographics as the source of misdirection.

Aging and growth. A growing literature studies how aging affects growth, with prevailing mechanisms on the supply side: labor scarcity (Maestas, Mullen, and Powell, 2023), reduced entrepreneurship (Liang, Wang, and Lazear, 2018), shifting saving patterns (Aksoy, Basso, and Smith, 2019), and induced automation (Acemoglu and Restrepo, 2017, 2022). We complement this work with a demand-side channel: aging shifts the composition of product demand, not just labor supply. Our channel is grounded in the empirical finding that older adults adopt new technologies more slowly than younger adults across a wide range of settings (Czaja and Sharit, 1998; Morris and Venkatesh, 2000; Czaja et al., 2006; Anderson and Perrin, 2017; Hauk, Hüfner, and Steinert, 2018). We embed this age-adoption friction in general equilibrium and show it is not merely distributional—it feeds back through product demand into firm innovation, with aggregate welfare consequences.

Dynamic inefficiency in OLG. The closest precedent is John and Pecchenino (1994), who show that environmental externalities generate dynamic inefficiency in OLG—the current generation degrades a common resource the next inherits. In our model, the "common resource" is the technology mix: current demand patterns erode retro-technology supply, imposing costs on future old consumers.

The rest of the paper is organized as follows. Section 2 provides the motivating evidence for the age-demand channel of innovation. Section 3 illustrates the simple theory that describes the age-demand channel. Section 4 presents the baseline model. Section 5 identifies the source of dynamic inefficiency, formalizes the structural decomposition (Proposition 5), and establishes the sensitivity of the externality to demographics. Section 6 provides the quantitative analysis, including calibration, transition dynamics, and the constrained-efficient policy. Section 7 concludes.

2 Motivating Facts and Empirical Analysis

This section establishes the empirical foundations of the model's central ingredients. We document three stylized facts: technology adoption rates vary systematically by age; market size directs technical change; and older consumers allocate a smaller share of their expenditure budget to new technology, even after controlling for income. The first two facts motivate the age-varying adoption cost Ψ and the market-size channel linking demographics to innovation incentives. The third — established using household-level expenditure data in Section 2.1 — provides direct micro-level evidence for the demand asymmetry at the heart of the model.

Technology Adoption Rates and Aging. The adoption cost, Ψ, is central to our model and reflects a well-documented empirical regularity. Younger populations adopt new technologies at higher rates, while older demographics exhibit inertia toward established ones (Weinberg, 2004). Crouzet et al. (2024) document mobile payment adoption in India, finding a strong negative correlation with age. Similarly, Meyer (2007) show that among German firms in ICT-intensive service sectors, those with a higher proportion of older workers were less likely to adopt new technologies. This suggests old consumers systematically face higher learning costs when adopting new technologies.

Market Size and Innovation. A large body of evidence establishes that the composition of demand shapes innovation, not merely its level. Focusing on exogenous changes driven by demographic trends in the U.S., Acemoglu and Linn (2004) find that a 1 percent increase in potential market size for a drug category leads to a 4–6 percent increase in the number of new drugs. This illustrates our model's central mechanism: market size directs innovation and shapes the economy's technology composition. We extend this logic: if young consumers constitute the dominant source of demand for new technologies, firms face stronger incentives to innovate in this direction. This is the age-demand channel of innovation—a tradition dating to Schmookler (1966).

Demographic Transition. Figure 1 documents the rise in old-age population shares across major economies since 1980. The share of the population aged 65 and over has risen substantially in all major economies, with Japan and Italy leading the transition. This demographic shift is the empirical backdrop against which our model's mechanism operates: as the share of old consumers grows, the technology-composition externality we identify becomes quantitatively more important.

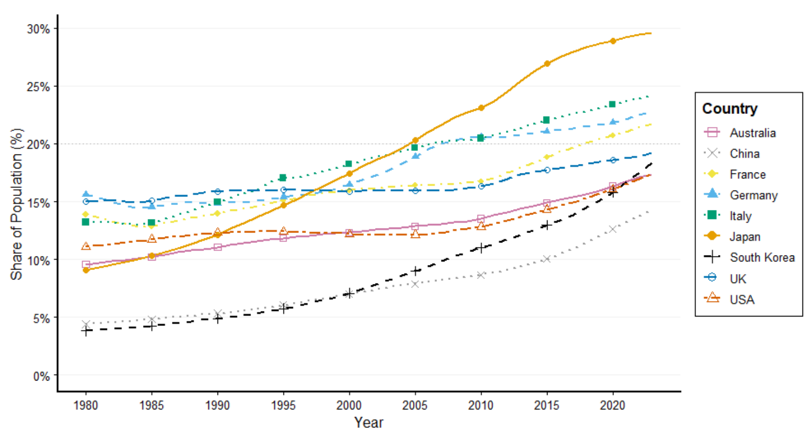


Figure 1: Demographic transition: population share aged 65+.

Notes: The figure shows the historical evolution of the old-age population share across major economies since 1980. Source: World Development Indicators (WDI), World Bank.

2.1 Age-based Demand Heterogeneity: Household Expenditure Data

The facts above establish the aggregate context. We now provide micro-evidence that older consumers systematically allocate a smaller share of their expenditure to new technologies, conditional on income – the demand-side asymmetry that is the model's central input.

Data. We use the US Consumer Expenditure Survey (CE), collected by the Bureau of Labor Statistics (BLS). The CE Interview Survey captures quarterly expenditure on larger recurring purchases, classified at the Universal Classification Code (UCC) level. We use data from 2024, covering Q1–Q4, yielding a cross-sectional sample of 12,184 consumer units.

Technology Classification. We construct a mapping from UCCs to the model's two technology types, new and retro, using the official BLS Hierarchical Groupings file (`CE-HG-Inter-2024.txt`). New-technology UCCs correspond to goods and services requiring active engagement with a digital interface, for which adoption costs are plausibly higher for older consumers; retro-technology UCCs are their incumbent alternatives. The full mapping is reported in Table 1.

Empirical Specification. For each household i, define the new- and retro-technology expenditure shares:

$$s_i^n = \frac{\sum_{k \in \mathcal{N}} e_{ik}}{\bar{e}_i}, \qquad s_i^r = \frac{\sum_{k \in \mathcal{R}} e_{ik}}{\bar{e}_i} \tag{1}$$

where $\mathcal{N}$ and $\mathcal{R}$ denote the sets of new- and retro-technology UCCs respectively, e_{ik} is household i's quarterly expenditure on UCC k, and $\bar{e}_i$ is total quarterly expenditure (`TOTEXPCQ`). These shares are the empirical counterparts to the model's consumption ratio $c_{O,n}/c_{Y,n}$. For each age group g, we report the weighted mean:

$$\bar{s}_g^j = \frac{\sum_{i \in g} w_i \cdot s_i^j}{\sum_{i \in g} w_i}, \qquad j \in \{n, r\} \tag{2}$$

where w_i denotes the CE final survey weight `FINLWT21`, ensuring estimates are representative of the US population rather than the raw sample. To assess whether the age gradient in s_i^j reflects age itself rather than income differences, we estimate the following weighted OLS regression for each $j \in \{n, r\}$:

$$s_i^j = \alpha + \sum_{g \in \mathcal{G}} \beta_g^j \cdot \mathbb{1}[\text{age}_i \in g] + \gamma \cdot \log(\text{income}_i) + \delta \cdot \text{family size}_i + \varepsilon_i \tag{3}$$

where $\mathcal{G} = \{[35, 49], [50, 64], [65, 79], [80+]\}$ and $[18, 34]$ is the omitted reference group. The regression is weighted by `FINLWT21`. The coefficients $\{\hat{\beta}_g^j\}$ measure the difference in expenditure share between age group g and the youngest group, conditional on log income and family size. Additional to this specification, we run two further specifications. Following Cravino, Levchenko, and Rojas (2022), we include income decile dummies, instead of $\log(\text{income})$, and then as a robustness check we include a third specification that includes regional dummies[1]. Regional dummies are included to control for geography – for example certain regions may have less access to technology.

Category	Description	UCC
New technology	Telephones and accessories (incl. smartphones)	320232
	Computers and computer hardware	690111
	Computer software	690119
	Computer accessories	690120
	Computer information services (internet)	690114
	Internet services away from home	690116
	Cellular phone service	270102
	Streaming and downloading audio	310350
	Streaming and downloading videos	310243
	Video game software	310231
	Video game hardware and accessories	310232
Retro technology	Cable and satellite television services	270310
	Residential telephone including VOIP	270106
	Televisions	310140
	VCRs and video disc players	310210
	Video cassettes, tapes, and discs	310220
	CDs, records, and audio tapes	310340
	Satellite radio services	270311

Table 1: UCC mapping to model technology types.

Notes: The table maps individual Universal Classification Codes (UCCs) in the U.S. Consumer Expenditure Survey to the model's two technology types. Source: BLS Hierarchical Groupings file, `CE-HG-Inter-2024.txt`.

Results. Figure 2 plots $\bar{s}_g^n$, $\bar{s}_g^r$, and mean quarterly spending levels by age group. The new-technology share declines monotonically from 10.4% for the 18–34 group

[1] Regions in the US CE Survey are broadly categorised as *Midwest*, *Northeast*, *South*, and *West*

to 6.9% for the 80+ group, while the retro-technology share rises, consistent with the adoption cost Ψ tilting old consumers' demand toward retro goods. In dollar terms, new-technology spending peaks at \$812 for the 35–49 group before falling sharply to \$307 for the 80+ group. Summary statistics are reported in Table 2.

Age group	Share (%)	Mean (\$)	Median (\$)	N
18–34	10.4	600	480	2,059
35–49	9.9	812	702	3,013
50–64	9.6	754	615	3,024
65–79	8.9	528	426	3,183
80+	6.9	307	239	905

Table 2: New-technology expenditure by age group.

Notes: The table reports the share, mean, and median of household quarterly expenditure on new-technology goods (UCC categories in Table 1) by household-head age group. Source: U.S. Consumer Expenditure Survey 2024; weighted using FINLWT21.

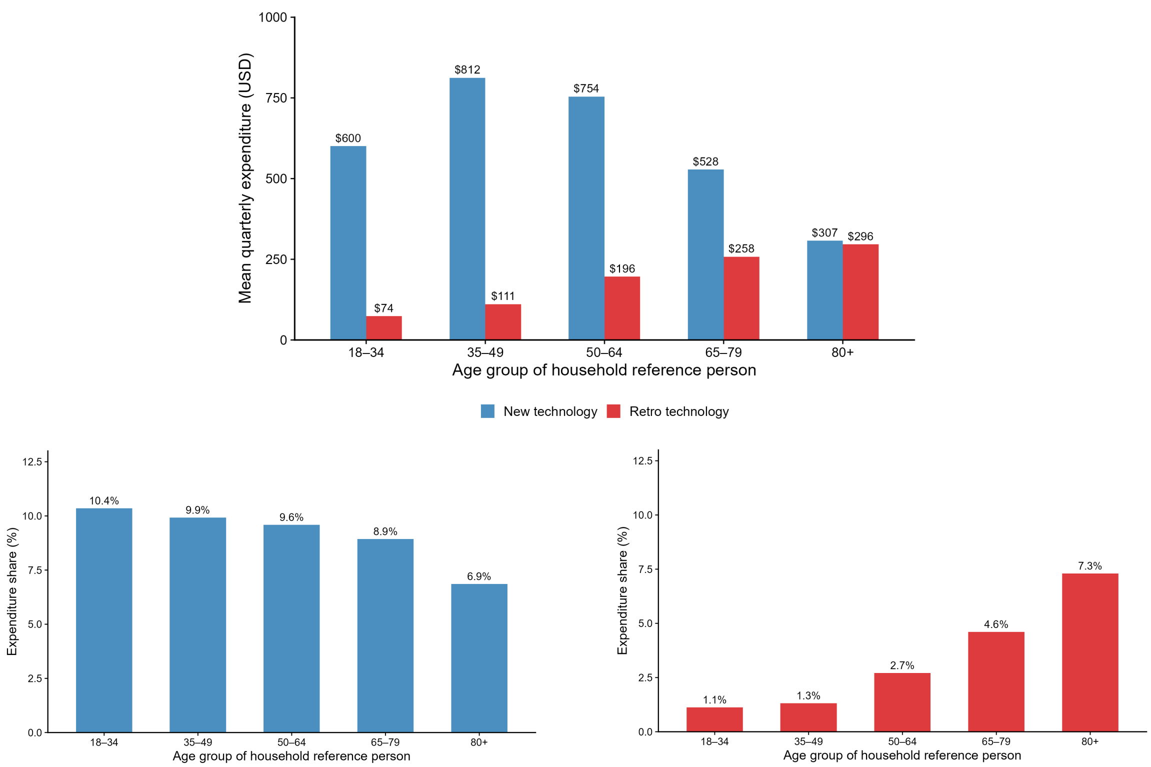


Figure 2: New- and retro-technology expenditure by age group.

Notes: Top: mean quarterly expenditure (USD) by technology type. *Bottom left*: new-technology share of total expenditure. *Bottom right*: retro-technology share of total expenditure. Source: U.S. Consumer Expenditure Survey 2024; weighted using FINLWT21.

Figure 3 reports the estimated coefficients $\{\hat{\beta}_g^n\}$ and $\{\hat{\beta}_g^r\}$ with 95% confidence intervals. The age gradient survives income controls and exhibits a striking asymmetry: the 35–49 and 50–64 coefficients are small and statistically indistinguishable from zero, while the 65–79 and 80+ groups show gaps of approximately 1.5 and 4 percentage points respectively in new-technology share, with mirror-image patterns in retro share. The concentration of the gap among retirees is consistent with an adoption cost that rises sharply upon exit from the labour market, when workplace exposure to new technology is lost, and is inconsistent with a pure income or cohort effect, which would predict a smooth gradient across all age groups rather than a break at retirement age.

Together, these findings establish that age-based demand heterogeneity in technology expenditure is a robust empirical regularity, not an artefact of income differences, and motivates the adoption cost $\Psi(c_n)$ at the heart of the model.

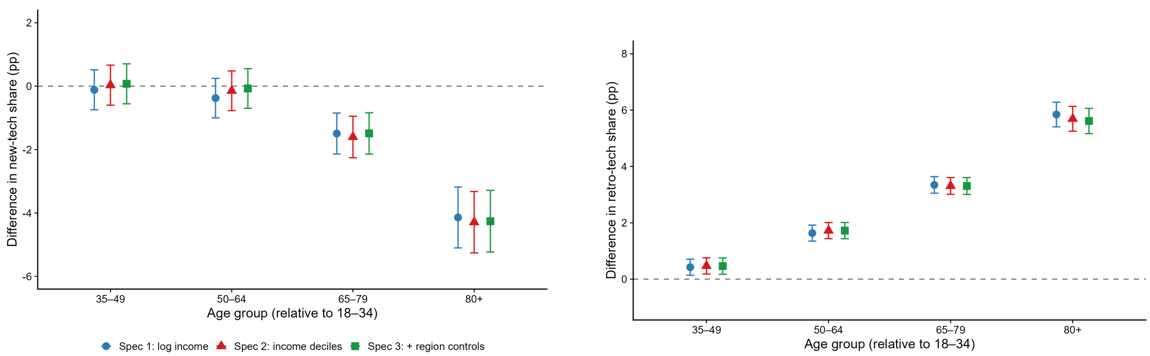


Figure 3: Age gradient in new- and retro-technology expenditure share.

Notes: Estimated coefficients $\{\hat{\beta}^n_g\}$ (left) and $\{\hat{\beta}^r_g\}$ (right) from equation (3) with 95% confidence intervals. All three specifications (log income, income decile dummies, income decile dummies plus regional dummies) are shown. Reference group: ages 18–34. Source: U.S. Consumer Expenditure Survey 2024.

3 Simple theory

We start with a simple illustration of a static economy in partial equilibrium, with demographics, adoption costs of new technology for old consumers and an innovating firm. This simple theory describes the demand-side channel, one of the core mechanisms of the paper. There are two types of good, $s \in \{n, r\}$. The focus here is on the firm's decision to innovate, hence we abstract from general equilibrium effects. For analytical tractability the adoption cost here is a *monetary* fee Ψ

that the old pay only when buying new-technology goods; the baseline model in Section 4 replaces it with a convex *utility* cost $\Psi(c_n) = (\psi/2)\, c_n^2$. Both formulations generate the same qualitative force—the old consume less new technology than the young—and the partial-equilibrium predictions below carry over to the general-equilibrium model.

Demographics. The economy is populated with a mass, normalised to unity, of consumers. Consumers are either *young* or *old*, of proportions ϕ_Y and ϕ_O respectively. The following holds:

$$\phi_Y + \phi_O = 1 \tag{4}$$

Households. Young and old households demand both technologies, retro and new. However, old households incur a fixed cost, Ψ, when consuming the new good. Households have preference share $\theta > 0.5$ for the new good and are endowed with Y. In partial equilibrium, the individual demand schedules are:

$$c^y(s) = \begin{cases} \frac{\theta Y}{P} & \text{for } \textit{new} \text{ good: } s = n \\ \frac{(1-\theta)Y}{P} & \text{for } \textit{retro} \text{ good: } s = r \end{cases} \tag{5}$$

$$c^o(s) = \begin{cases} \frac{\theta Y - \Psi}{P} & \text{for } \textit{new} \text{ good: } s = n \\ \frac{(1-\theta)Y}{P} & \text{for } \textit{retro} \text{ good: } s = r \end{cases} \tag{6}$$

Aggregate Demand. We aggregate individuals' demands for the two types of good, accounting for the demographic structure:

$$C(s, \phi) = \phi_Y c^y(s) + \phi_o c^o(s) \tag{7}$$

Firms. There is a single profit-maximizing firm, which can produce either the retro technology, r, or new technology type, n, but not both. If the firm chooses to produce the new technology type, n, it must choose innovation effort, e; with probability $p(e)$, the firm operates the new technology, and with probability $1 - p(e)$, it operates the old technology.

The firm faces the aggregate demand schedule for both types of technology, and takes prices as given. Prices, $\{P, w\}$, are exogenous with $w = 0$. The profits from operating both technologies are the following:

$$\pi(s,\phi) = (P - w)C(s,\phi) = P \cdot C(s,\phi) \tag{8}$$

$$\pi(s,\phi) = \begin{cases} \phi_Y \theta Y + \phi_O(\theta Y - \Psi) & \text{for } \textit{new} \text{ good: } s = n \\ (1-\theta)Y & \text{for } \textit{retro} \text{ good: } s = r \end{cases} \tag{9}$$

The firm value, J, is:

$$J(\phi) = \max\{J^e(\phi), J^r(\phi)\} \tag{10}$$

$$J^e(\phi) = \max_{e \geq 0} -\Phi(e,\phi) + p(e)[\pi(n,\phi)] + (1 - p(e))[\pi(r,\phi)] \tag{11}$$

$$J^r(\phi) = \pi(r,\phi) \tag{12}$$

Firm Optimality. There are two key conceptual points. The first is that $J^e(\phi)$ is monotonically increasing in ϕ_Y. This is due to the increased profit incentive from producing the new good. The adoption cost reduces old demand for the new good. Given $\theta > 0.5$, a rising proportion of young consumers increases the aggregate demand for the new good. Conditional on innovating, which is determined by the extensive margin, such that $J^e(\phi) > J^r(\phi)$, the firm chooses innovation effort e that satisfies:

$$\Phi_e(e^*,\phi) = p_e(e^*)[\pi(n,\phi) - \pi(r,\phi)] \tag{13}$$

Where subscript e denotes the partial derivative with respect to e.

The second is that innovation intensity, e^*, is increasing in ϕ_Y. For simplicity, under linear cost, and $p(e) = \frac{e}{1+e}$, we can derive the following:

$$\pi(n,\phi) - \pi(r,\phi) = \Delta\pi = Y(2\theta - 1) - \Psi(1 - \phi_Y) \tag{14}$$

$$\frac{\partial \Delta\pi}{\partial \phi_Y} = \Psi > 0 \tag{15}$$

Then solving for e^*, we have:

$$e^* = \sqrt{\frac{\Delta\pi}{\psi(\phi)}} - 1 \tag{16}$$

And so we have innovation intensity increasing in ϕ_Y, by the following:

$$\frac{\partial e^*}{\partial \phi_Y} = \underbrace{\frac{\partial \Delta\pi}{\partial \phi_Y}}_{>0} \underbrace{\frac{\partial e^*}{\partial \Delta\pi}}_{>0} > 0 \tag{17}$$

Theoretical Predictions. Under linear innovation cost and neutral demographic proficiency ($\psi(\phi) = 1$), the model yields the following predictions:

Proposition 1. *There exists a threshold level ϕ^* of the proportion of young such that $J^e(\phi^*) = J^r(\phi^*)$. For $\phi > \phi^*$, the firm opts to innovate ($e^* > 0$). This is the extensive margin of the age-demand channel of innovation.*

Proof. $J^e(\phi)$ is monotonically increasing in ϕ (the increased young share raises new-good profits via $\theta > 1/2$ and the adoption-cost asymmetry), while $J^r(\phi)$ is independent of ϕ. Existence of the threshold follows from the intermediate value theorem. ∎

Proposition 2. *Innovation intensity e^* is increasing in ϕ, the proportion of young consumers. This is the intensive margin of the age-demand channel of innovation.*

Proof. The optimality condition gives $e^* = \sqrt{\Delta\pi/\psi} - 1$, with $\Delta\pi = Y(2\theta - 1) - \Psi(1 - \phi_Y)$ increasing in ϕ_Y. ∎

Proposition 3. *The probability of a successful innovation, $p(e^*)$, is increasing in ϕ. The rate at which innovations arrive in the economy thus rises with a larger young share.*

Proof. $\dfrac{\partial p(e^*(\phi))}{\partial \phi} = \underbrace{\dfrac{\partial e^*(\phi)}{\partial \phi}}_{>0 \text{ by Proposition 2}} \underbrace{\dfrac{\partial p}{\partial e}}_{>0 \text{ for } e \geq 0} > 0.$ ∎

Proposition 4. *The overall effect of demographics on innovation through both supply and demand channels is ambiguous; it depends on how demographics affect the proficiency of the innovation process. In a 'young improve proficiency' scenario the threshold ϕ^* falls because innovating is cheaper; in a 'young raise the cost of innovation' scenario the threshold rises and innovation may collapse altogether.*

Proof. Direct, by signing $\partial e^*/\partial\phi$ and $\partial\phi^*/\partial\phi$ under the alternative specifications of $\psi(\phi)$ in the cost function. ■

4 Baseline Model

Time is discrete with an infinite horizon. The economy consists of overlapping generations of households and a unit continuum of firms. The central question is how demographic composition—the relative sizes of young and old cohorts—shapes firms' incentive to innovate.

4.1 Demographics

Households live for two periods: *young* (Y) and *old* (O). Each period, a new cohort of young households of size ϕ_0 is born, where

$$\phi_0 \sim \Gamma. \tag{18}$$

Let ϕ_Y and ϕ_O denote the measures of young and old households. The demographic state evolves as:

$$\phi_O' = \phi_Y, \tag{19}$$
$$\phi_Y' = \phi_0. \tag{20}$$

The current young become next period's old, and a new cohort arrives. Variation in cohort sizes $\{\phi_0\}$ is the sole source of aggregate uncertainty.

Notation: cohort masses, shares, and aging. We use ϕ_Y and ϕ_O throughout to denote cohort *masses*, not population shares. The corresponding old-population share is $s_O \equiv \phi_O/(\phi_Y + \phi_O)$. Two distinct comparative-statics exercises use these objects in different ways and should not be conflated. (i) "Aging" along the calibrated transition (Section 6.2) refers to a permanent decline in the inflow ϕ_0, so that ϕ_Y falls first and ϕ_O follows with one-period lag, raising s_O. (ii) "Symmetric demographic state ϕ" in the cross-state sensitivity sweep (Section 5.3) sets $\phi_Y = \phi_O = \phi$, so $s_O = 0.5$ is held fixed and ϕ varies the *total* population per cohort. Increasing ϕ in (ii) is therefore a market-size expansion, not aging. The innovation-cost scaling $\Phi(e) = (\kappa/(2\phi_O))\, e^2$ depends on the mass ϕ_O, capturing a mentor/managerial-experience channel: more old workers in absolute numbers cheapens innovation regardless of the old share. We use the ϕ-sweep to isolate the technology-composition externality from market-size effects, and the transition to isolate aging proper.

4.2 Aggregate State

The aggregate state is:

$$X = \big(Q_n,\, \phi_O,\, \phi_Y\big), \tag{21}$$

where Q_n is the measure of new-technology firms (defined below) and (ϕ_O, ϕ_Y) describes the demographic structure.

4.3 Households

All households consume two types of goods: *new-technology* goods (n) and *retro-technology* goods (r). Within-period preferences are Cobb–Douglas:

$$u(c_n, c_r) = \theta \log c_n + (1-\theta) \log c_r, \quad \theta \in (0,1), \tag{22}$$

where c_s denotes consumption of type-s goods for $s \in \{n, r\}$.

Old household. An old household enters the period holding a shares of the aggregate firm portfolio and consumes all wealth:

$$v_O(a; X) = \max_{c_n, c_r} \quad \theta \log c_n + (1-\theta) \log c_r - \Psi(c_n) \tag{23}$$

$$\text{s.t.} \quad p_n\, c_n + p_r\, c_r = \big(D(X) + H(X)\big)\, a, \tag{24}$$

where $p_s = P_s$ is the type-s price index in symmetric equilibrium, $D(X)$ is aggregate dividends, and $H(X)$ is the ex-dividend market value of the firm portfolio. The term $\Psi(c_n)$ is an *age-related adoption cost*: a disutility that old households suffer when consuming new-technology goods. We assume Ψ is increasing and convex with $\Psi(0) = 0$; specifically, $\Psi(c_n) = \frac{\psi}{2}c_n^2$ with $\psi > 0$.

The adoption cost Ψ is central to the model. It captures frictions that older consumers face when using new-technology goods—the time spent learning an unfamiliar interface, the effort of switching from established routines, the cognitive burden of adapting to products designed for younger users—without constituting a monetary transfer. Young households face no such cost. This generational asymmetry creates a wedge in marginal utility: at an interior optimum, the old household's first-order condition is $\theta/c_n - \psi\, c_n = \lambda\, p_n$, where λ is the marginal utility of wealth. The adoption cost tilts the old's demand toward retro-technology goods, so an economy with more old consumers directs less spending toward new technology. As we show in Section 5, this demand-side asymmetry feeds back to the supply side, shaping the profitability of innovation and the economy's technology composition.

Young household. A young household earns wage income $w(X)$ and has no initial assets. It saves by purchasing shares of the firm portfolio:

$$v_Y(X) = \max_{c_n, c_r, a'} \quad \theta \log c_n + (1-\theta) \log c_r + \beta\, \mathbb{E}\big[v_O(a'; X') \,\big|\, X\big] \tag{25}$$

$$\text{s.t.} \quad p_n\, c_n + p_r\, c_r + a'\, H(X) = w(X), \tag{26}$$

where a' is share holdings carried into the next period and $\beta \in (0, 1)$ is the discount factor. The expectation is over next period's cohort size $\phi_0' \sim \Gamma$.

Optimal demands. The Cobb–Douglas structure yields closed-form demands. For the young, who face no adoption cost:

$$c_{Y,n} = \frac{\theta\, W_Y}{p_n}, \quad c_{Y,r} = \frac{(1-\theta)\, W_Y}{p_r}, \quad \text{where } W_Y = w - a'\, H(X). \tag{27}$$

For the old, the adoption cost $\Psi(c_n) = (\psi/2)c_n^2$ distorts demand away from new-technology goods. The first-order conditions are:

$$\frac{\theta}{c_{O,n}} - \psi\, c_{O,n} = \lambda_O\, p_n, \quad \frac{1-\theta}{c_{O,r}} = \lambda_O\, p_r, \tag{28}$$

where λ_O is the marginal utility of old-age wealth $W_O = (D+H)\,a$. Eliminating λ_O and substituting the budget constraint $p_r\,c_{O,r} = W_O - p_n\,c_{O,n}$ yields a single equation in $c_{O,n}$:

$$\left(\frac{\theta}{c_{O,n}} - \psi\,c_{O,n}\right)(W_O - p_n\,c_{O,n}) = (1-\theta)\,p_n. \tag{29}$$

When $\psi = 0$ this reduces to the standard Cobb–Douglas solution $c_{O,n} = \theta\,W_O/p_n$. For $\psi > 0$, (29) is a cubic in $c_{O,n}$ with a unique interior solution in $(0, \theta\,W_O/p_n)$, reflecting how the adoption cost tilts old consumption toward retro goods.

4.4 Firms

There is a unit continuum of firms. Each firm's current technology type is $s \in \{n, r\}$, where n denotes new technology and r denotes retro technology. Let Q_s be the measure of type-s firms, with $Q_n + Q_r = 1$.

Production and profit. Firms within each technology type produce differentiated varieties under monopolistic competition with within-type elasticity of substitution $\varepsilon > 1$. Each household's consumption of type-s goods aggregates over varieties:

$$c_s = \left(\int_0^{Q_s} c_s(j)^{(\varepsilon-1)/\varepsilon}\,dj\right)^{\varepsilon/(\varepsilon-1)}, \tag{30}$$

where $c_s(j)$ denotes consumption of variety j of type s. In symmetric equilibrium, the type-s price index is $P_s = Q_s^{1/(1-\varepsilon)}\mu\,w$, where $\mu = \varepsilon/(\varepsilon-1)$ is the constant markup and w is the wage. Each type-s firm earns per-firm profit:

$$\pi(s; X) = \frac{w}{\varepsilon-1}\,C_s\,Q_s^{-\varepsilon/(\varepsilon-1)}, \tag{31}$$

where $C_s = \phi_O\,c_{O,s} + \phi_Y\,c_{Y,s}$ is aggregate demand for type-s goods. Production requires labor at constant marginal cost w, which serves as the numeraire ($w = 1$).

Innovation and firm value. At the start of each period, a firm inherits a past technology type $s = s_{t-1}$. Within the period, it chooses innovation effort $e \geq 0$

at cost $\Phi(e; X)$ and the resulting technology is realized stochastically: with probability $p(s, e)$ the firm operates new technology *this period*, and with probability $1 - p(s, e)$ it operates retro technology. R&D is therefore a within-period gamble whose payoff is realized in the current period; the inherited type only matters as the state determining the success probability. We adopt the following functional forms:

$$p(s, e) = \bar{p}_s + (1 - \bar{p}_s)\,\frac{e}{1 + e}, \quad \bar{p}_s \in [0, 1), \tag{32}$$

$$\Phi(e; X) = \frac{\kappa}{2\,\phi_O}\,e^2, \quad \kappa > 0. \tag{33}$$

The transition probability $p(s, e)$ is bounded in $[\bar{p}_s, 1)$: the base rate $\bar{p}_s$ is the probability of operating new technology this period even without innovation effort, and effort drives p toward unity with diminishing returns. The marginal probability $p_e(s, e) = (1 - \bar{p}_s)/(1 + e)^2$ is decreasing in e, so that the first unit of effort is the most productive. The innovation cost Φ is quadratic in effort and decreasing in the old population ϕ_O, reflecting the role of experienced workers as mentors and managers in R&D.[2]

The value of a firm with inherited type s at the start of the period satisfies:

$$\begin{aligned} J(s; X) = \max_{e \geq 0} \quad & - \Phi(e; X) \\ & + p(s, e)\,\Big[\pi(n; X) + \mathbb{E}\big[q(X, X')\,J(n; X') \bigm| X\big]\Big] \\ & + \big(1 - p(s, e)\big)\,\Big[\pi(r; X) + \mathbb{E}\big[q(X, X')\,J(r; X') \bigm| X\big]\Big], \end{aligned} \tag{34}$$

so that today's profit $\pi(\cdot; X)$ is realized at the within-period type, not at the inherited type s. The inherited type enters only through the success probability $p(s, e)$ and the cost $\Phi(e; X)$. where $q(X, X')$ is the stochastic discount factor. The young household's Euler equation for share purchases pins down q:

$$q(X, X') = \beta\,\frac{\lambda_O(X')}{\lambda_Y(X)}, \tag{35}$$

[2]The $1/\phi_O$ scaling ensures that per-firm innovation costs fall when a larger old cohort provides a deeper pool of experienced labor for R&D. This creates an additional channel through which demographics affect innovation: aging not only shifts demand composition but also reduces the cost of innovation.

where $\lambda_Y = 1/W_Y$ and $\lambda_O = (1-\theta)/(P_r\, c_{O,r})$ are the marginal utilities of wealth for the young and old households, and $W_Y = w - a' H(X)$ and $W_O = (D(X) + H(X))\, a$ denote their respective total expenditures.

4.5 Aggregation and Market Clearing

Goods market. Aggregate demand for type-s goods at price p is the population-weighted sum across generations:

$$C(p, s; X) = \phi_O\, c_O(s; p, X) + \phi_Y\, c_Y(s; p, X), \tag{36}$$

where $c_i(s; p, X)$ is per-capita demand for type-s goods by generation $i \in \{O, Y\}$.

Asset market. The aggregate firm portfolio is in unit supply. Old households enter the period holding the entire portfolio; young households purchase it by the end of the period. Asset market clearing requires:

$$\phi_O\, a = 1 \quad \Longrightarrow \quad a = \frac{1}{\phi_O}, \tag{37}$$

$$\phi_Y\, a' = 1 \quad \Longrightarrow \quad a' = \frac{1}{\phi_Y}. \tag{38}$$

Labor market. Only the young supply labor. Each firm demands labor for production and innovation. In symmetric equilibrium, labor market clearing requires:

$$\phi_Y = \underbrace{\sum_{s\in\{n,r\}} Q_s\, \ell(s; X)}_{\text{production labor}} + \underbrace{\sum_{s\in\{n,r\}} Q_s\, \Phi\big(e(s; X); X\big)/w}_{\text{innovation labor}}, \tag{39}$$

where $\ell(s; X) = C_s\, Q_s^{-\varepsilon/(\varepsilon-1)}$ is per-firm production labor demand and $\Phi(e; X)/w$ converts the per-firm innovation cost (a wage-bill in the numeraire) into labor units. Since $w = 1$ is numeraire, this condition pins down aggregate output.

Dividends and market value. Aggregate dividends equal net profits after innovation expenditure:

$$D(X) = \sum_{s\in\{n,r\}} \Big(\pi(s;X) - \Phi\big(e(s;X);X\big)\Big)\, Q_s, \tag{40}$$

where $e(s;X)$ is the optimal innovation effort of type-s firms.

The ex-dividend market value of the firm portfolio is:

$$\begin{aligned} H(X) = \sum_{s\in\{n,r\}} Q_s \Big[& p\big(s, e(s;X)\big)\, \mathbb{E}\big[q(X,X')\, J(n;X') \mid X\big] \\ & + \Big(1 - p\big(s, e(s;X)\big)\Big)\, \mathbb{E}\big[q(X,X')\, J(r;X') \mid X\big]\Big]. \end{aligned} \tag{41}$$

Note that $D(X) + H(X) = \sum_s Q_s\, J(s;X)$: total firm value equals dividends plus ex-dividend market value.

Within-period realization of the technology composition. Let $Q_n^- = Q_n(t-1)$ denote the inherited (past) composition at the start of period t. By the law of large numbers, the realized composition within period t is deterministic:

$$\begin{aligned} Q_n &= p(n,e_n)\, Q_n^- + p(r,e_r)\,(1-Q_n^-) \\ &= \underbrace{p_{nn}}_{\text{success of n-firms}} Q_n^- + \underbrace{p_{rn}}_{\text{catch-up of r-firms}} (1-Q_n^-), \end{aligned} \tag{42}$$

where $p_{nn} \equiv p(n,e_n)$ is the probability that an inherited n-firm successfully operates new technology this period, and $p_{rn} \equiv p(r,e_r)$ is the probability that an inherited r-firm catches up to new technology this period. In a stationary equilibrium, $Q_n = Q_n^-$ requires $Q_n(1-p_{nn}) = p_{rn}(1-Q_n)$: the within-period flow out of n equals the within-period flow in from r. Note the timing: today's realized Q_n is determined by today's effort given the inherited state Q_n^-, so the corrective tax in period t acts on within-period innovation rather than on a future realization.

4.6 Recursive Competitive Equilibrium

Definition 1. *A* ***recursive competitive equilibrium*** *consists of:*

1. *value functions:* $v_O(a; X)$, $v_Y(X)$, *and* $J(s; X)$*;*

2. *household policy functions: consumption* $\{c_i^s\}_{s\in\{n,r\}}$ *for each generation* $i \in \{O, Y\}$ *and savings* $a'(X)$*;*

3. *firm policy functions: pricing* $p(s; X)$ *and innovation effort* $e(s; X)$*;*

4. *aggregate objects: goods prices* $\{p_s(X)\}_s$*, wage* $w(X)$*, dividends* $D(X)$*, market value* $H(X)$*, and stochastic discount factor* $q(X, X')$*;*

5. *a law of motion* $X' = G(X, \phi_0)$*;*

such that:

1. ***Household optimality.*** v_O *and* v_Y *solve* (23)–(24) *and* (25)–(26)*, with demands given by* (27) *and* (28)–(29)*.*

2. ***Firm optimality.*** J *satisfies* (34) *with innovation FOC* (45)*, and prices solve* (31)*.*

3. ***Goods market clearing.*** *Aggregate demand* (36) *equals supply for each* $s \in \{n, r\}$*.*

4. ***Asset market clearing.*** *Conditions* (37)–(38) *hold.*

5. ***Labor market clearing.*** *Condition* (39) *holds.*

6. ***Pricing consistency.*** *The SDF* $q(X, X')$ *satisfies* (35)*.*

7. ***Laws of motion.*** *The aggregate state evolves according to* (42)*,* (19)*, and* (20)*.*

5 Source of Dynamic Inefficiency

The competitive equilibrium features a *technology-composition externality*: the decentralized allocation of innovation does not reflect the full social costs and benefits of the technology mix. Two forces drive the market toward over-innovation—the standard *variety externality* from monopolistic competition, operating through the OLG structure, and the *adoption cost externality* specific to aging. Even without adoption costs, the OLG structure creates a wedge between private and social incentives; adoption costs amplify this wedge.

5.1 The Demand Externality

When the young choose their consumption bundle $(c_{Y,n}, c_{Y,r})$, they equate the marginal rate of substitution to the price ratio:

$$\frac{\theta}{1-\theta} \cdot \frac{c_{Y,r}}{c_{Y,n}} = \frac{p_n}{p_r}. \tag{43}$$

This condition treats prices as given. In general equilibrium, the young's demand feeds back into the realized technology mix through three channels. First, young households face no adoption cost, so their demand tilts aggregate spending toward new-technology goods. Second, higher new-technology demand raises $\pi(n; X) - \pi(r; X)$ and the value gap $J(n; X) - J(r; X)$, raising innovation effort e within the period. Third, higher e raises the realized Q_n via (42), shifting today's technology mix toward new technology—which becomes the inherited state Q_n^- entering tomorrow. The young do not internalize this chain. A higher Q_n reduces retro-technology availability today and rolls forward as Q_n^- tomorrow—precisely when the same household, now old, faces a larger adoption-cost-bearing share of consumption.

The externality is compounded on the firm side. Under within-period realization, the innovation decision is governed by:

$$\Phi_e(e; X) = p_e(s, e)\Big[\big(\pi(n; X) - \pi(r; X)\big) + \mathbb{E}\big[q(X, X')\,(J(n; X') - J(r; X'))\,\big|\,X\big]\Big], \tag{44}$$

where Φ_e and p_e denote partial derivatives with respect to e. Substituting the functional forms (32)–(33), the FOC becomes:

$$\frac{\kappa}{\phi_O}\, e = \frac{1-\bar{p}_s}{(1+e)^2}\,\big[\Delta\pi + q\,\Delta J\big], \tag{45}$$

where $\Delta\pi \equiv \pi(n; X) - \pi(r; X)$ and $\Delta J \equiv \mathbb{E}[J(n; X') - J(r; X') \mid X]$. The left side is the marginal cost of effort; the right side is the marginal benefit—the probability gain per unit of effort times the sum of the contemporaneous profit gap (because the realized type determines this period's profit) and the discounted continuation-value gap. Both firm types face the same gap $\Delta\pi + q\,\Delta J$, so the equal marginal cost condition $\Phi'(e_n)/p_e(n, e_n) = \Phi'(e_r)/p_e(r, e_r)$ pins down relative innovation intensities across types.

Two features of (45) amplify the inefficiency. First, the stochastic discount factor $q(X, X') = \beta \lambda'_O / \lambda_Y$—where λ_Y and λ'_O are marginal utilities of wealth for today's young and tomorrow's old—is set by the young household's Euler equation for share purchases. It does not reflect old welfare directly. Second, both the contemporaneous profit gap $\Delta\pi$ and the continuation-value gap ΔJ reflect private profitability at prevailing prices, not the social value of the technology mix.

What firms internalize, and what they do not. The over-innovation result does not arise from a forecasting failure. Firms correctly anticipate that today's young become tomorrow's old and that new-technology profitability evolves accordingly. The value gap $J(n; X') - J(r; X')$ in (44) incorporates all of this. What firms do not internalize is their collective effect on prices. Each firm's innovation marginally shifts Q_n, which moves relative prices (P_n down, P_r up) and pushes old households toward costlier new-technology goods. No atomistic firm prices this welfare cost, because it operates through the aggregate technology composition. The planner corrects this by choosing a lower Q_n^*—not from better forecasts, but from internalizing the price-index burden on old consumers.

A missing priced cost for the technology choice. The externality is best understood as arising from a *missing price for the technology choice itself*. The model has two intertwined missing prices, and they reinforce each other.

(i) No factor-market signal between technologies. Both technology types use the same labor at the same wage and earn the same within-type markup; there is no specialized capital, no industry-specific human capital, and no input whose relative price would adjust as Q_n shifts. The only signal of relative scarcity between new- and retro-technology production runs through the goods-market price index, $P_s = Q_s^{1/(1-\varepsilon)} \mu w$, which depends on the aggregate composition Q_n rather than on any input quantity that an atomistic firm would purchase. With technology-specific factors—R&D-heavy capital for new-technology firms, generalist labor for retro firms—a rising Q_n would bid up the price of the new-technology factor, internally signaling scarcity to entrants and partially correcting the externality. Stripping that channel out by symmetrizing inputs gives our externality its full bite: the only remaining feedback runs through the price-index burden on old consumers,

which atomistic firms do not internalize. The mechanism is therefore most relevant for innovation in domains where the two technologies share inputs—services, software, and consumer-facing products fit this description—and is attenuated where new-technology production draws on specialized, scarce factors.

(ii) No market for the aggregate technology composition. Even if one allowed for technology-specific factors and let input prices adjust, no market explicitly prices Q_n as an aggregate state. In the competitive equilibrium, the marginal benefit of innovation that each firm sees is the private gap $\Delta\pi + q\,\Delta J$—contemporaneous profit and continuation-value at *prevailing* prices. There is no security or futures contract through which a firm pays for, or is compensated for, its marginal contribution to the aggregate Q_n. If firms had to purchase a "new-technology license" priced at the full social marginal cost of raising Q_n—including the welfare loss old households suffer through the price-index channel—they would internalize the externality. The corrective tax τ that emerges in the constrained-efficient analysis below is exactly this missing price: a Pigouvian wedge that the planner attaches to innovation effort to make firms face the social marginal cost of the technology choice they are de facto making.

These two missing prices have a clean economic interpretation. The first says that there is no *factor-market* channel through which the choice between new and retro technology gets priced; the second says that there is no *financial-market* channel through which the choice gets priced either. The technology-composition externality is what is left once both have been ruled out: a wedge that exists because the choice variable—which technology to operate—enters everyone's budget constraints through the price index but does not appear as a transactable quantity in any market. A pecuniary externality of this form is welfare-neutral under complete markets (Arrow and Debreu, 1954); it is welfare-relevant here because the OLG structure makes markets incomplete in three specific ways: old households cannot contract over the future technology mix; the SDF $q = \beta\lambda_O'/\lambda_Y$ is pinned by the young household's Euler equation and does not directly price old-household welfare; and Q_n is a public good with no market to allocate it (Greenwald and Stiglitz, 1986; Geanakoplos and Polemarchakis, 1986). In the language of Geanakoplos and Polemarchakis (1986), the corrective tax *completes the market* by internalizing the general-equilibrium feedback from Q_n to prices to old-household welfare. The vertical gap between social and private marginal cost in Figure 4 measures exactly this missing price.

5.2 What drives the externality? A structural decomposition

The discussion above identifies the technology-composition externality with a missing price for the technology choice. The externality is not a generic feature of OLG models with monopolistic competition: it requires two structural conditions—a non-trivial composition margin and demand heterogeneity—amplified by a third, shared inputs. We state this as a proposition, then walk through each ingredient in turn.

Proposition 5 (Structural decomposition of the externality)**.** *In the OLG economy of Section 4, the technology-composition wedge $Q_n^{CE} - Q_n^* > 0$ requires conditions (A) and (B) below; condition (C) governs its magnitude.*

(A) ***Multiple technologies.*** *There are at least two technology types and $Q_n \in (0, 1)$ is endogenously chosen by atomistic firms.*

(B) ***Heterogeneous demand.*** *The marginal social valuation of a new-technology firm differs across consumer types; formally, $\partial U_i / \partial Q_n \neq \partial U_j / \partial Q_n$ for some types $i \neq j$ at the prevailing CE prices.*

(C) ***Shared inputs.*** *Both technology types are produced from a common factor pool at common factor prices ($\partial w / \partial Q_n = 0$ from the firms' perspective).*

Let one ingredient be replaced by its structural alternative:

(i) Single-technology economy (A fails). The composition margin does not exist; the wedge is undefined. (A) is necessary.

(ii) Homogeneous demand (B fails). The firm's value gap $\Delta\pi + q\,\Delta J$ coincides with the welfare-weighted social valuation; the wedge collapses to zero. (B) is necessary and sufficient under (A).

(iii) Disjoint inputs (degenerate failure of C). Each technology uses one technology-specific factor exclusively, so Q_n is pinned by the relative factor endowments and

is no longer a free firm choice. The wedge is trivially zero; this case collapses (C) into a stronger version of (A) failing.[3]

(iv) Differentiated inputs (non-degenerate failure of C). Both technologies use multiple factors but in different intensities; factor prices respond to Q_n. Under (B), the wedge shrinks but generically remains positive: factor prices internalize the supply-side dimension, but the demand-side wedge from (B) is uncorrected. Under homogeneous demand, the wedge is zero. (C) is an amplifier, not necessary: removing (C) alone in this analytically interesting form does not eliminate the wedge.

The Pigouvian tax τ implementing Q_n^* is the missing price for the technology choice. (B) is the binding ingredient—its removal eliminates the wedge regardless of factor structure. Case (iii) suppresses the wedge by closing the composition margin (a degenerate case); case (iv) is the analytically interesting one, in which factor prices internalize the supply-side dimension but the demand-side wedge from (B) survives.

Proof. See Online Appendix Section A. ■

Numerical illustration along (B): the ψ-sweep. We can illustrate the role of (B) directly using the comparative statics already implemented in our calibration: vary the adoption-cost parameter ψ and recompute the static-CEP wedge and corrective tax at the calibrated demographic point ($\phi = 1$).

The $\psi = 0$ row corresponds to the case where the preference channel of (B) is shut off (old and young have identical preferences). The wedge does not vanish: the OLG structure itself generates budget heterogeneity (cohort-specific wealth profiles interacting with the price-index channel), which suffices to keep (B) active. About four-fifths of the baseline wedge is driven by the OLG-budget channel of heterogeneity; the residual fifth is the preference channel.[4]

[3] Strictly, $(A) \wedge$ (iii) is empty: under truly disjoint inputs an atomistic firm cannot switch technology types because it would need a factor it does not have access to, so Q_n is no longer endogenously chosen by atomistic firms and condition (A) fails. We list this case separately because the proximate mechanism—extreme factor specialization—is conceptually distinct from the absence of a second technology, even though it closes the composition margin in the same way. The substantive content of the proposition is therefore that the wedge requires (A) and (B); (C) governs the magnitude only when the composition margin is open, i.e., when (C) is either fully shared or weakly differentiated rather than disjoint.

[4] To shut off (B) entirely would require collapsing the OLG to a representative agent, which is a different model. We therefore obtain only a partial decomposition along (B); the full first-row

Table 3: The role of demand heterogeneity in the technology-composition wedge.

ψ	Gap $Q_n^*/Q_n^{CE} - 1$	Implied τ	Interpretation
0	-6.1%	$+55\%$	Preference heterogeneity off; only OLG-budget heterogeneity remains.
3.14 (baseline)	-7.6%	$+72\%$	Calibrated combination of preference and budget heterogeneity.
9 (high)	-9.5%	$+97\%$	Extreme preference heterogeneity.

Notes: Each row reports the static-CEP relative gap $Q_n^*/Q_n^{CE} - 1$ (negative, since CE over-innovates: $Q_n^{CE} > Q_n^*$) and the implied innovation tax τ at the calibrated demographic point ($\phi = 1$), holding all other parameters at the baseline calibration of Table 4 and varying only the adoption-cost parameter ψ. The $\psi = 0$ row shuts off the preference channel of heterogeneous demand; budget heterogeneity from the OLG structure remains active. Source: `agetech_cep.m` comparative-statics output.

Ingredient (C): factor-market completeness. The role of (C) is conceptual rather than numerical in our analysis. The argument is straightforward: factor prices price *factor scarcity*, while the welfare loss old households suffer through the price-index channel depends on *adoption costs and consumer heterogeneity*. These are unrelated quantities, so the factor-market response does not generically match the welfare-side response. Adding technology-specific factors would therefore partially internalize the supply-side dimension but leave a demand-side residual under (B). The mechanism we identify is therefore most relevant when the two technologies share inputs—services, software, and consumer-facing products fit this description—and is attenuated where new-technology production draws on specialized, scarce factors.

Take-away. The technology-composition externality requires *two* things together: a non-trivial composition margin (A) and demand heterogeneity (B). Shared inputs (C) amplify the externality but are not strictly necessary—under differentiated inputs (C$\to$2), the wedge shrinks because factor prices partially internalize the supply-side dimension, but the demand-side wedge from (B) survives. Heterogeneous demand is therefore the binding ingredient: without it the externality is gone regardless of factor structure. This is why our externality is precisely calibrated to demographic aging: aging makes demand heterogeneous in a particular,

"zero-wedge under homogeneous demand" claim is supported by the proof sketch above rather than by a numerical exercise.

strong, age-correlated way. In an economy where consumer demand were homogeneous in the relevant sense the same demographic shift would have no welfare consequences for the technology-composition margin; specializing factors (without homogenizing demand) would attenuate but not eliminate the externality.

Relation to existing literature. The structural decomposition above echoes the directed-technical-change literature (Acemoglu, 2002; Acemoglu et al., 2012), in which a multiple-technology composition margin (A) interacts with a structural reason that consumers value the two technologies differently to generate biased innovation. Two pieces of our framing differ. First, the demand-heterogeneity ingredient (B) is age-driven in our setting, generated by the OLG structure plus the adoption cost ψ, rather than by skill or factor intensity as in the standard directed-technical-change setup. Second, we identify (B) rather than (C) as the binding ingredient: in our setup with shared inputs, removing (B) eliminates the wedge regardless of factor structure, while removing (C) only attenuates it. The demand-driven-innovation tradition (Schmookler, 1966; Foellmi and Zweimüller, 2006) provides the (A)+(B) structure but does not formalize (C); we make (C) explicit and clarify its role as an amplifier. The missing-market interpretation in §5 builds on the pecuniary-externalities tradition (Greenwald and Stiglitz, 1986; Geanakoplos and Polemarchakis, 1986), and the corrective tax τ plays the role of the price that "completes" the market for the technology choice.

5.3 Sensitivity of the Externality to Demographics

To understand how demographics shape the externality, we decompose the social marginal cost of innovation. At the CE effort level, the private and social marginal costs are:

$$\mathrm{MC}(e) = \frac{\kappa}{\phi_O} e, \tag{46}$$

$$\mathrm{SMC}(e) = (1+\tau)\,\frac{\kappa}{\phi_O} e, \tag{47}$$

where $\tau > 0$ is the innovation tax that implements Q_n^*. The externality wedge is:

$$\mathrm{SMC}(e) - \mathrm{MC}(e) = \tau \cdot \frac{\kappa}{\phi_O} e. \tag{48}$$

Two opposing forces govern how the wedge responds to demographics. The cost factor $\kappa e/\phi_O$ falls with population size, as a larger cohort reduces the per-capita cost of innovation. But the corrective tax τ rises, because more consumers are affected by the technology-mix distortion, amplifying the aggregate welfare cost that the planner internalizes.

We evaluate this decomposition numerically by solving both the CE and CEP at each symmetric demographic state $\phi_O = \phi_Y = \phi \in [0.8, 1.5]$, computing MC, τ, and SMC at the planner allocation (so τ and MC are both evaluated at the same effort, yielding a clean wedge $\mathrm{SMC} - \mathrm{MC} = \tau \cdot \mathrm{MC}$). Figure 5 reports the results.

Sensitivity of the externality wedge. Across symmetric demographic states $\phi_O = \phi_Y = \phi \in [0.8, 1.5]$:

(i) The private marginal cost $\mathrm{MC} = \kappa e/\phi$ is monotonically decreasing in ϕ, falling from 0.012 at $\phi = 0.8$ to 0.010 at $\phi = 1.5$.

(ii) The corrective tax τ is increasing in ϕ, rising from approximately 69% to 83%.

(iii) The proportional wedge $\mathrm{SMC}/\mathrm{MC} = 1 + \tau$ is increasing in ϕ, rising from 1.69 to 1.83.

(iv) The absolute wedge $\mathrm{SMC} - \mathrm{MC}$ is approximately constant near 0.008 across all ϕ (mean 0.0082, range $[0.0081, 0.0083]$).

The two forces in (48) nearly exactly offset: τ rises with ϕ at roughly the same rate that $\kappa e/\phi$ falls, leaving the absolute wedge stable. But the *proportional* wedge—the ratio of social to private cost—rises substantially. In a larger economy, innovation is cheaper per capita, but the social cost of the technology-mix distortion is an increasing multiple of that private cost.

This result has two implications. First, the externality is quantitatively large and robust: the corrective tax τ ranges from 69% to 83% across all demographic states, confirming that the technology-composition distortion is not a small or fragile wedge. Second, the near-constancy of the absolute wedge means that the over-innovation gap $e^{CE} - e^{SP}$ is *persistent* across demographic regimes. As the economy ages, private costs rise (higher $\kappa e/\phi_O$), reducing equilibrium innovation effort—but the planner would reduce it by the same absolute amount. The market's demographic adjustment is in the right direction but insufficient in magnitude, regardless of the population's age structure.

Figure 4 illustrates this graphically. Aging shifts both the MC and SMC curves upward (higher $\kappa e/\phi_O$), while the marginal benefit curve shifts down (smaller young cohort reduces the value gap). The CE effort falls, but the vertical gap between SMC and MC is preserved, so the over-innovation gap persists.

Demographics and innovation intensity. The same comparative static reveals how demographics shape the *level* of innovation. Figure 6 panel (a) plots the CE innovation effort for new-technology and retro-technology firms across ϕ. Both e_n and e_r rise monotonically with population size: at $\phi = 0.8$ (an aging economy), $e_r = 0.120$ and $e_n = 0.038$; at $\phi = 1.5$ (a younger economy), $e_r = 0.172$ and $e_n = 0.058$—an increase of roughly 44% and 52%, respectively. Retro firms innovate substantially more than new-technology firms ($e_r \approx 3\,e_n$), reflecting the asymmetry in base transition rates: retro firms must "catch up" from a low base persistence $\bar{p}_r = 0.05$, whereas new-technology firms already enjoy high persistence $\bar{p}_n = 0.74$. Over the calibrated demographic transition (ϕ: $1.30 \to 1.00$), total innovation effort $e_n + e_r$ falls from 0.215 to 0.184—a 14% decline, or roughly 5% per 0.1 drop in ϕ.

The mechanism is the innovation first-order condition (45): the marginal cost of effort $\kappa e/\phi$ falls with population size while the marginal benefit—the discounted value gap $q(J'_n - J'_r)$—rises, because a larger economy generates higher aggregate demand, widening profit differentials between firm types. An economy with a larger, younger population thus innovates more intensely. This connects directly to the cross-country evidence motivating the paper: countries with younger demographics—such as the United States or India—should exhibit higher innovation intensities than rapidly aging economies like Japan or South Korea, holding other factors constant.

Panel (b) shows the implied technology composition. Both Q_n^{CE} and Q_n^* rise with ϕ, confirming that younger economies sustain a larger share of new-technology firms. The gap between CE and planner is persistent: the market always over-innovates, with $Q_n^*/Q_n^{CE} - 1$ around -8% at low ϕ and -6% at high

ϕ. The over-innovation wedge *narrows* as the economy grows younger, consistent with the adoption-cost externality being most severe when old consumers are a large share of the population.

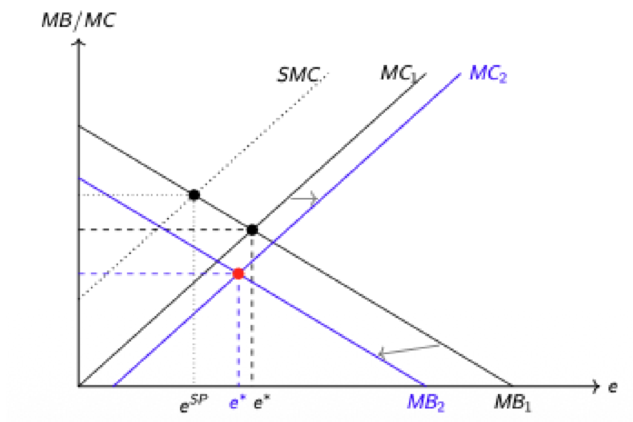


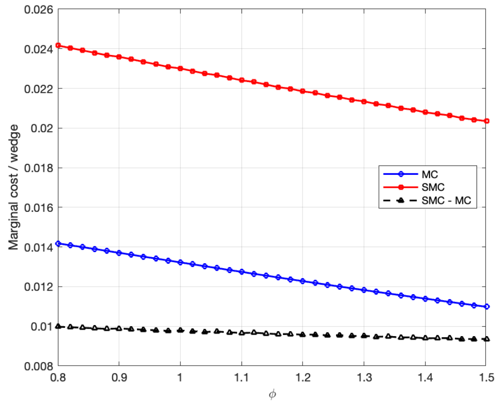
Figure 4: Innovation effort: private vs. social marginal cost.

Notes: The upward-sloping curves are the private marginal cost of effort MC $= \kappa e/\phi_O$ (black: baseline; blue: after aging, when higher ϕ_O lowers MC). The downward-sloping curves are the marginal benefit MB $= p_e(s,e)\, q\, (J_n' - J_r')$ (black: baseline; blue: after aging, when the value gap shrinks). The dashed SMC curve adds to private MC the marginal welfare cost that innovation imposes on old households through the price-index channel. The vertical gap between SMC and MC is preserved across demographic states. The CE equilibrium e^* ($\mathrm{MC}_1 \cap \mathrm{MB}_1$) exceeds the planner's optimum e^{SP} ($\mathrm{SMC} \cap \mathrm{MB}_1$). Aging reduces effort to e°, but the over-innovation gap persists.

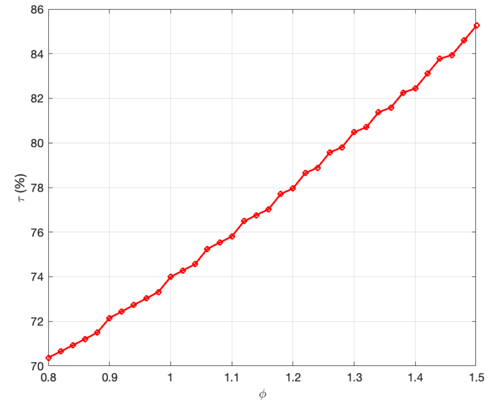
5.4 Constrained-Efficient Technology Policy

To isolate the technology-composition externality from intergenerational redistribution, we study a *constrained-efficient* allocation. The planner controls only the innovation margin—choosing the stationary share of new-technology firms Q_n via a proportional subsidy or tax on innovation costs—while all other equilibrium mechanisms remain as in the CE. Households optimize given prices, asset markets clear, and the wealth distribution is determined by portfolio returns.

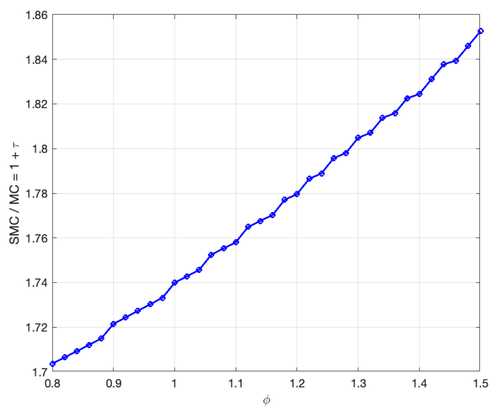
For each candidate Q_n, the constrained planner solves for the CE allocation conditional on that technology composition, subject to stationarity and the inherited equal marginal cost condition from firm optimality. The planner evaluates contemporaneous utilitarian welfare $\mathcal{W}(Q_n) = u(c_Y) + \tilde{u}(c_O)$, where $\tilde{u}$ includes



(a) MC, SMC, and wedge



(b) Corrective tax τ



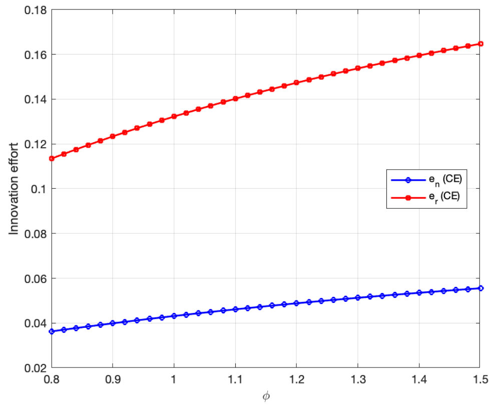
(c) Proportional wedge SMC/MC $= 1 + \tau$

Figure 5: Sensitivity of the externality to demographics.

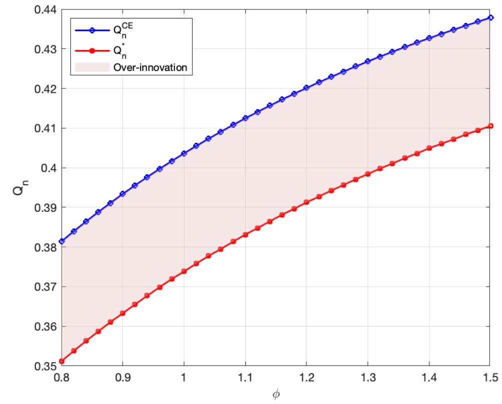
Notes: Each point solves the CE and CEP at symmetric $\phi_O = \phi_Y = \phi$. Panel (a): private marginal cost (MC, blue), social marginal cost (SMC, red), and the absolute externality wedge (dashed black). Both MC and SMC fall with ϕ, but the wedge is approximately constant. Panel (b): the corrective tax τ rises smoothly from 69% to 83% as ϕ increases, reflecting the larger aggregate welfare cost of the technology-mix distortion in bigger economies. Panel (c): the proportional wedge $1 + \tau$ rises from 1.69 to 1.83, confirming that the social cost of innovation is an increasing multiple of private cost.

the adoption cost, and chooses $Q_n^* = \arg\max \mathcal{W}(Q_n)$.[5]

[5] The criterion sums utility across cohorts *alive in the same period*, with no β weighting between the young and the old. This is the same criterion used throughout the quantitative analysis of Section 6 (transition welfare in Figure 7 panel (d) and the dynamic CEP in Section 6.4). A generation-discounted lifetime utilitarian criterion would weight the old's utility by β; given $\beta = 0.96^{25} \approx 0.36$ at the calibrated annual rate, this would shift the planner's preference toward young consumption but not eliminate the technology-composition externality, which arises from the price-index channel rather than the welfare weights.



(a) Innovation effort



(b) Technology composition: CE vs. planner

Figure 6: Demographics, innovation intensity, and the over-innovation gap.
Notes: Each point solves the CE and CEP at symmetric $\phi_O = \phi_Y = \phi$. Panel (a): innovation effort by firm type. Both e_n (new-tech, blue) and e_r (retro, red) rise monotonically with ϕ. Retro firms innovate roughly three times as intensely as new-tech firms, reflecting the lower base persistence $\bar{p}_r$. Aging (lower ϕ) depresses innovation across both firm types. Panel (b): technology composition in the CE (blue) and planner's allocation (red). Both Q_n^{CE} and Q_n^* rise with ϕ, but the market always over-innovates. The gap between the two curves narrows as ϕ increases, confirming that the adoption-cost externality is most severe in aging economies.

The externality. The constrained planner reveals that the CE over-innovates. At the baseline calibration ($\varepsilon = 50$, $\psi = 3.14$): $Q_n^* \approx 0.370 < Q_n^{CE} \approx 0.400$, a gap of roughly -7.6%. Two forces contribute to this gap. The first is the standard *variety externality* from monopolistic competition: firms' innovation changes the price indices $P_s \propto Q_s^{1/(1-\varepsilon)}$, affecting consumer surplus. The second is the *adoption cost externality*: the price shift from higher Q_n pushes old consumers toward goods that are costly for them to adopt. Both operate through the price-index channel that no atomistic firm internalizes. Even at $\psi = 0$, the planner prefers fewer new-technology firms (-6.1% gap), reflecting the OLG structure's inherent wedge between private innovation incentives and social welfare. As ψ rises, the gap widens further—to -11.4% at $\psi = 20$—confirming that the adoption cost amplifies the inefficiency.

Table 4: Calibration

Parameter	Value	Target / Source
Panel A: Externally set		
Discount factor, β	0.3604	Annual 4% $\Rightarrow 0.96^{25}$
CES elasticity, ε	50	Near-homogeneous within-type varieties
Expenditure share, θ	0.50	Symmetric (normalization)
Innovation cost, κ	0.10	R&D spending $\approx$ 3% of dividends
Panel B: Internally calibrated (moment matching)		
Adoption cost, ψ	3.14	$c_{O,n}/c_{Y,n} = 0.35$; Czaja et al. (2006)
Base persistence, $\bar{p}_n$	0.7387	$p(n, e_n) = 0.75$; Akcigit and Kerr (2018)
Base catch-up, $\bar{p}_r$	0.0510	$Q_n = 0.40$; OECD (2020)

Notes: One model period is 25 years. Panel A reports parameters set externally; Panel B reports the three parameters jointly calibrated to match the three moments listed in the right column. The calibration achieves an exact match on all three moments. The realized R&D-to-dividends ratio at the baseline is approximately 3.1%, close to the 3% target.

6 Quantitative analysis

6.1 Calibration

Table 4 summarizes the parameter values. One model period is 25 years. Three parameters—β, ε, θ—are set externally: $\beta = 0.96^{25} \approx 0.3604$ implies a standard annual discount rate of 4%; $\varepsilon = 50$ yields a within-type markup of 2%, ensuring that within-type varieties are close substitutes; $\theta = 0.50$ is a symmetric normalization. The innovation cost scale $\kappa = 0.10$ is informally chosen so that per-period R&D costs represent approximately 3% of aggregate dividends; the realized moment is verified in the calibration output.

The remaining three parameters—the adoption cost ψ, the base persistence rate $\bar{p}_n$, and the base catch-up rate $\bar{p}_r$—are jointly calibrated to match three moments. The old-to-young new-technology consumption ratio $c_{O,n}/c_{Y,n} = 0.35$ reflects the technology usage gap between older and younger adults documented by Czaja et al. (2006). Technology persistence over a 25-year horizon $p(n, e_n) = 0.75$ is consistent with patent-based measures of frontier persistence in Akcigit and Kerr (2018). The new-technology firm share $Q_n = 0.40$ targets the share of firms that have adopted advanced digital technologies, following OECD (2020). The calibration achieves an exact match for all three moments.

6.2 Transition Dynamics

We study the economy's perfect foresight transition as the young cohort size ϕ_Y declines from $\phi_{\text{high}} = 1.30$ (a younger society, representing the 1930s) to $\phi_{\text{low}} = 1.00$ (the calibrated terminal steady state). One period equals 25 years. The old population follows with a one-period lag: $\phi_O(t) = \phi_Y(t-1)$. This asymmetry—the young shrink before the old do—is the source of transitional dynamics beyond a simple shift between steady states. We use $T = 15$ periods so that the system has enough room to converge to the terminal steady state at the boundary; the demographic shift itself, parameterized as a logistic centered at $t = 7$, is essentially complete by $t \approx 11$, and the late periods serve as the terminal-condition cushion required by the backward-forward algorithm.

Algorithm. We compute the transition path using backward-forward iteration on the aggregate price sequences $\{q(t), V(t), Q_n(t)\}_{t=1}^{T}$, where $T = 15$.[6]

Technology composition. Figure 7 reports the transition paths. As the young cohort shrinks, aggregate demand shifts away from new-technology goods: fewer young consumers means less spending on goods for which they face no adoption cost. New-technology firms' profits fall relative to retro firms', reducing the value gap $J(n) - J(r)$ and hence innovation incentives. The new-technology share Q_n declines from 0.424 in the initial steady state to 0.400 in the terminal steady state, a decline of approximately 5.6%. The decline is gradual and S-shaped, mirroring the logistic path of ϕ_Y.

Innovation effort. The decline in Q_n is driven by falling innovation effort. Both e_n and e_r decline during the transition (panel (c)), as the shrinking young population compresses the profit gap between firm types. The innovation first-order condition $\Phi'(e) = p'(s,e)\, q\, [J(n; t+1) - J(r; t+1)]$ confirms that a lower value gap $J(n) - J(r)$ reduces the marginal return to innovation.

[6] The backward pass solves firm values, innovation effort, and household consumption from $t = T$ to $t = 1$, taking terminal steady-state values as the boundary condition. The forward pass simulates the law of motion for Q_n from the initial condition. A dampened update brings the guessed and computed paths into alignment; the iteration converges to a tolerance of 10^{-8}.

Welfare: the transition is borne by the old. The welfare consequences are starkly asymmetric across generations. Panel (d) shows that aggregate welfare $W(t) = u_Y(t) + u_O(t)$ passes through a pronounced trough, bottoming out at periods 7–8. Panel (e) reveals who bears the cost: the intergenerational welfare gap $u_Y - u_O$ spikes from approximately 1.02 in both steady states to 1.10 at the transition peak. Young households' utility barely moves—they consume whatever technology the market offers without friction. Old households, by contrast, suffer a sharp welfare decline.

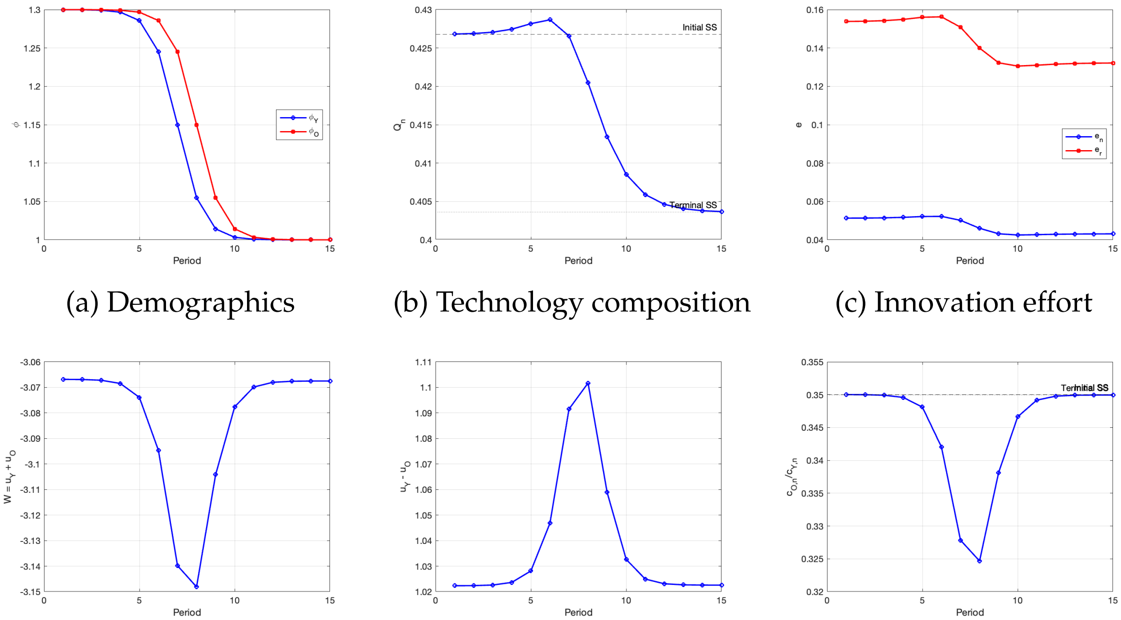
The mechanism is the one-period demographic lag: $\phi_O(t) = \phi_Y(t-1)$. At the steepest part of the transition, the young cohort has already shrunk but the old cohort—yesterday's large young—is still large. The old inherit a technology mix shaped by a previously larger young cohort, rich in new-technology firms. The technology mix adjusts too slowly, and the old bear the cost.

Stochastic discount factor. These welfare dynamics are mirrored in asset prices. The stochastic discount factor $q(t) = \beta\,\lambda_O(t+1)/\lambda_Y(t)$ spikes during the transition, rising from 0.93 to above 1.0 at the peak. Today's large young cohort will sell shares to a smaller future young cohort, depressing the future share price. That q exceeds unity—a negative real interest rate—signals dynamic inefficiency driven by the demographic transition itself.

Consumption. Finally, the old-to-young new-technology consumption ratio $c_{O,n}/c_{Y,n}$ (panel (f)) is approximately 0.35 at both steady states (matching the calibration target) and dips to about 0.32 at the transition peak, when the price index P_n is most adverse for old consumers. The temporary widening of the consumption gap mirrors the welfare trough in panel (d) and the spike in $u_Y - u_O$ in panel (e).

6.3 Welfare Analysis: CE vs. Constrained-Efficient Planner

The constrained-efficient planner of Section 5.4 prescribes a technology composition Q_n^* that differs from the CE allocation Q_n^{CE} at every demographic level. To assess how the externality evolves during the transition, we solve the planner's problem at each symmetric demographic level $\phi \in [1.0, 1.3]$ and interpolate the



(a) Demographics (b) Technology composition (c) Innovation effort

(d) Aggregate welfare (e) Welfare gap: $u_Y - u_O$ (f) New-tech C ratio

Figure 7: Perfect foresight transition path.

Notes: Perfect-foresight transition path as the young cohort declines from $\phi_Y = 1.30$ to $\phi_Y = 1.00$. Each period represents 25 years. Panel (d) plots aggregate welfare $W(t) = u_Y(t) + u_O(t)$, which exhibits a pronounced trough during the transition. Panel (e) plots the intergenerational gap $u_Y - u_O$, which spikes as old households bear the brunt of the externality.

resulting $Q_n^*(\phi)$ schedule along the transition's demographic path $\phi_Y(t)$.[7]

The over-innovation wedge widens during the transition. Figure 8 plots the CE and planner technology compositions along the transition, together with the externality gap $Q_n^*/Q_n^{CE} - 1$. At the terminal steady state ($\phi = 1.0$), the gap is approximately -7.6%. At the initial steady state ($\phi = 1.3$), the gap is approximately -6.8%. The gap *widens* to -10.2% at the peak of the demographic transition (periods 7–8), precisely when ϕ_Y is declining most rapidly.

The widening has a clear economic interpretation. As the young cohort shrinks, the CE technology composition Q_n^{CE} falls—but not fast enough. The planner, internalizing the adoption cost externality, would reduce Q_n more aggressively, recog-

[7]This $Q_n^*(\phi_Y(t))$ schedule is therefore an approximation: the static planner is solved at $\phi_Y = \phi_O = \phi$, while along the actual transition ϕ_Y and ϕ_O differ by one period. We index the interpolation on ϕ_Y because the young's demand drives innovation incentives via the SDF $q = \beta \lambda_O' / \lambda_Y$ and ϕ_Y pins down current labor supply. The full non-stationary planner—solved jointly across the transition—is the dynamic CEP of Section 6.4.

nizing that the growing old share of the population suffers disproportionately from new-technology abundance. The market, by contrast, adjusts Q_n only through the indirect channel of reduced new-technology profitability, which responds sluggishly to demographic change.

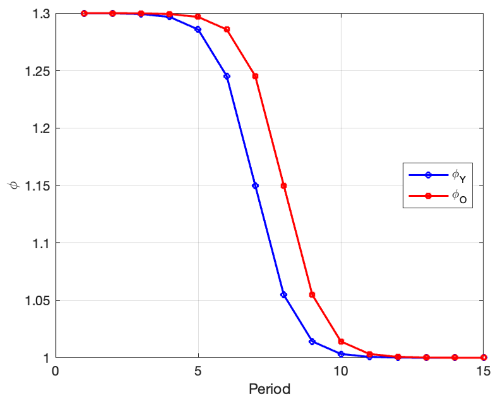
The welfare cost of the externality rises during the transition. The intergenerational welfare gap $u_Y(t) - u_O(t)$ spikes from 1.02 in both steady states to 1.10 at the peak (periods 7–8). Old households bear the brunt: the technology mix adjusts sluggishly while the mismatch between the shrinking young and still-large old cohort is most acute. Panel (d) of Figure 8 confirms this, plotting welfare under the CE and the planner along the transition. The CE welfare drops faster; the gap widens during periods 6–10, mirroring the over-innovation wedge in panel (c).

The schedule $Q_n^*(\phi)$. Figure 9 plots $Q_n^{CE}(\phi)$ and $Q_n^*(\phi)$ as functions of the demographic parameter ϕ. Both are increasing in ϕ: a younger society supports more new-technology firms in both the CE and the planner's allocation, because more young consumers expand new-technology demand. However, Q_n^* rises more steeply than Q_n^{CE}, so the gap between them narrows as the economy becomes younger. The externality is largest in an old society and smallest in a young one—aging amplifies the inefficiency.

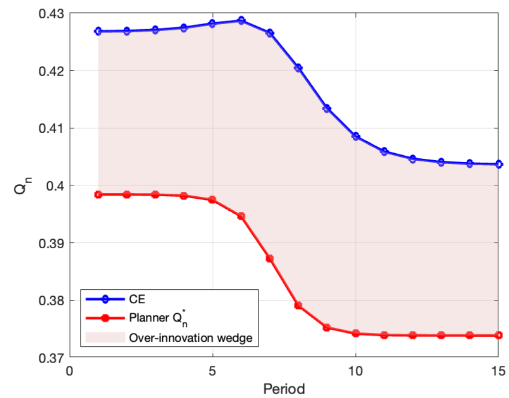
Implied innovation tax. The static-CEP innovation tax τ that implements Q_n^* at each demographic level along the transition ranges from approximately 72% to 78%. The tax is highest at the transition's start, when ϕ_Y is at its initial high level. The fully dynamic, Markov-perfect counterpart in Section 6.4 is somewhat lower (about 60%–63%), reflecting that the dynamic planner accounts for inherited Q_n^- when choosing today's Q_n, which moderates the required wedge. Either way, the corrective policy is not a fixed levy but a time-varying instrument indexed to the economy's demographic state.

6.4 Dynamic Constrained-Efficient Policy: Markov-Perfect

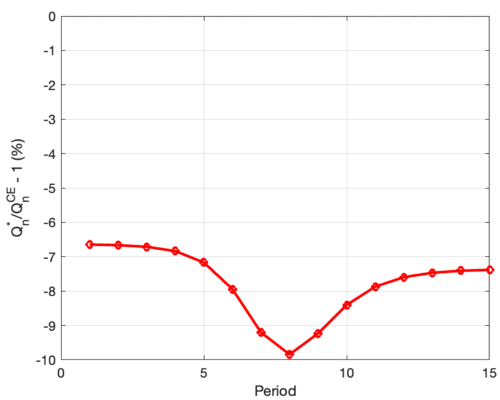
The static CEP of Section 5.4 evaluates welfare at each demographic level independently, treating the technology composition as a sequence of steady states.



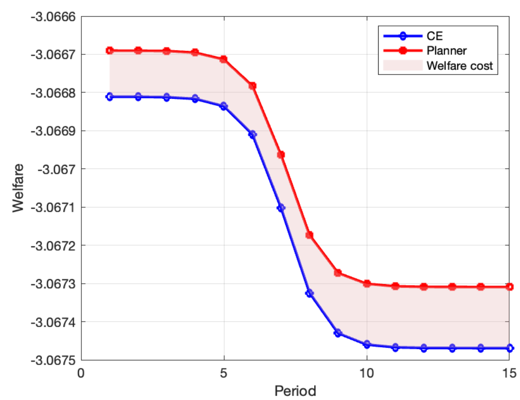
(a) Demographics



(b) Technology composition



(c) Externality gap



(d) Welfare: CE vs. Planner

Figure 8: Competitive equilibrium vs. constrained-efficient planner along the transition.

Notes: Panel (b) shows $Q_n^{CE}(t)$ (blue) and $Q_n^*(t)$ (red), with the shaded area indicating the over-innovation wedge. Panel (c) plots the gap $Q_n^*/Q_n^{CE} - 1$ (percent). Panel (d) plots welfare under the CE (blue) and the planner (red); the shaded area is the welfare cost of the externality, which visibly widens during the demographic transition.

This misses intertemporal linkages: firm values $J(s;t)$ depend on future values $J(s;t+1)$, and the stochastic discount factor $q(t)$ links generations across time. We now compute the dynamic constrained-efficient policy under the within-period realization timing of Section 4, formulated as the on-path realization of a Markov-perfect (time-consistent) equilibrium. We solve for the on-path policy via iterative best-response on the deterministic transition path; we do not solve the value function over the full (Q_n^-, ϕ_t) state space. Proposition 7 establishes that, on a deterministic perfect-foresight trajectory, the iterative-best-response fixed point coincides

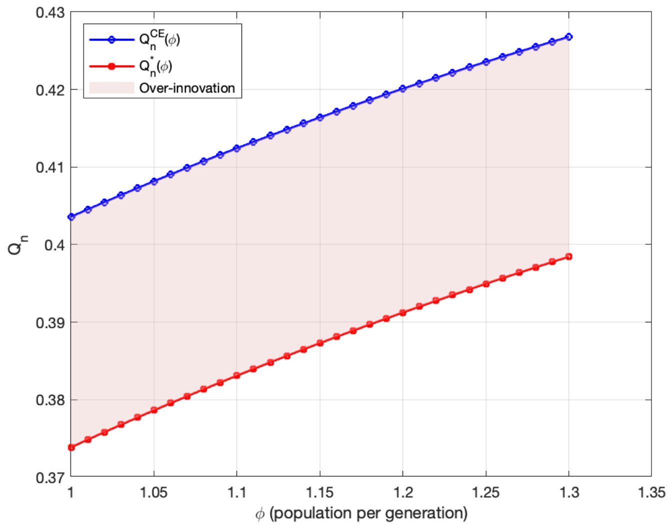


Figure 9: Technology composition as a function of the demographic parameter ϕ.

Notes: ϕ is the population per generation, evaluated at symmetric demographics $\phi_O = \phi_Y = \phi$. Blue: $Q_n^{CE}(\phi)$. Red: $Q_n^*(\phi)$. The shaded area is the over-innovation wedge. Both increase in ϕ, but the CE rises faster, so the gap widens as the population ages (lower ϕ).

with the on-path realization of the MPE policy.

Bellman formulation. At the start of period t, the relevant state is the pair (Q_n^-, ϕ_t), where $Q_n^- \equiv Q_n(t-1)$ is the inherited technology stock and $\phi_t \equiv (\phi_Y(t), \phi_O(t))$ is the demographic state evolving deterministically along the calibrated transition. The planner's choice variable is the current technology composition $Q_n \in [0,1]$. Given Q_n, the within-period CE block of Section 4—household optimization, firm pricing, asset-market clearing, labor-market clearing—pins down period-t flow utilities $u_Y(Q_n; Q_n^-, \phi_t)$ and $u_O(Q_n; Q_n^-, \phi_t)$ and the LoM $Q_n^- \to Q_n$ as the state for $t+1$. Following the welfare criterion of Section 5.4, we adopt utilitarian flow welfare $u(Q_n; Q_n^-, \phi_t) \equiv u_Y + u_O$ with no inter-temporal discounting across cohorts (a standard utilitarian criterion in OLG; see footnote 5). The planner's value function $W(Q_n^-, \phi_t; t)$ satisfies the Bellman equation

$$W(Q_n^-, \phi_t; t) \;=\; \max_{Q_n \in [0,1]} \Big\{ u(Q_n; Q_n^-, \phi_t) \;+\; W(Q_n, \phi_{t+1}; t+1) \Big\}, \qquad (49)$$

with the terminal condition $W(\cdot, \phi_T; T) = u(\cdot; \cdot, \phi_T)$ at the end of the demographic transition (after which ϕ is constant and the problem is autonomous).

Markov-perfect equilibrium. A *Markov-perfect equilibrium* is a policy $Q_n^*(\cdot, \cdot; t)$ such that, for every state (Q_n^-, ϕ_t) and every t, $Q_n^*(Q_n^-, \phi_t; t)$ achieves the maximum in (49) when the continuation $W(\cdot, \phi_{t+1}; t+1)$ is the value function induced by the same policy from t+1 onward. The MPE is time-consistent by construction: the planner's t-th choice remains optimal after t has elapsed, with no commitment device required.

Trajectory iteration: reconciling the local and global objectives. On the deterministic perfect-foresight transition, we solve the MPE by iterative best response on the on-path sequence $\{Q_n^*(t)\}_{t=1}^T$. The following lemma justifies the local one-period objective used inside each iteration.

Proposition 6 (Local problem reduces to two-flow welfare)**.** *Fix an iterate $\{Q_n^{it}(s)\}_{s \neq t}$ and consider the planner's choice of $Q_n(t)$, holding $\{Q_n^{it}(s)\}_{s \neq t}$ fixed. Then*

$$\sum_{s=1}^{T} u\big(Q_n(s); Q_n(s-1), \phi_s\big) \; = u\big(Q_n(t); Q_n^{it}(t-1), \phi_t\big) + u\big(Q_n^{it}(t+1); Q_n(t), \phi_{t+1}\big) + C_t, \tag{50}$$

where C_t collects all terms not involving $Q_n(t)$. Hence the period-t best-response problem reduces to maximizing $u(t) + u(t+1)$ alone.

The proof is immediate: $Q_n(t)$ enters period-t flow utility through the choice itself (it is the contemporaneous Q realized within t) and period-t+1 flow utility as the inherited state Q_n^-. It does not enter any other period's flow under the LoM $Q_n^-(s) = Q_n(s-1)$ holding $\{Q_n^{\text{it}}(s)\}_{s \neq t}$ fixed, because $Q_n(s)$ for $s \neq t$ is fixed at the iterate and $Q_n^-(s) = Q_n^{\text{it}}(s-1)$ is constant for $s \neq t, t+1$. The equivalence between maximizing the un-discounted continuation $\sum_{s \geq t} u(s)$ and the two-flow local objective is therefore exact, not approximate, at each iteration on a deterministic trajectory.

Proposition 7 (Convergence to MPE)**.** *Suppose the period-t flow welfare $u(\cdot; Q_n^-, \phi_t)$ is strictly concave on $[0,1]$ for every (Q_n^-, ϕ_t) encountered along the calibrated trajectory. Then the iterative best-response on the on-path sequence converges to a fixed point $\{Q_n^*(t)\}_{t=1}^T$ that satisfies the Bellman (49) along the realized state path, and is therefore the on-path realization of a Markov-perfect equilibrium policy.*

At our calibration, the strict-concavity hypothesis is verified numerically along the iterates by inspection of the second-order term $\partial^2 u/\partial Q_n^2 < 0$ at every grid point. The iteration is therefore well-defined and converges; in practice we observe a contraction rate of roughly 0.4–0.6 per outer iteration in the calibrated run.

Architecture. Each outer iteration performs (i) a backward pass along the current trajectory to compute aggregate paths $(q(t), V(t), \lambda_O(t))$ consistent with rational expectations of the iterate's future, and (ii) a forward solve, period by period, of the local problem (50). The implied innovation efforts $(e_n(t), e_r(t))$ are recovered from the LoM and the equal-marginal-cost condition $\Phi'(e_n)/p_e(n, e_n) = \Phi'(e_r)/p_e(r, e_r)$, which ensures a single proportional tax $\tau(t)$ implements both firm types. The tax is recovered from the firm FOC under within-period realization,

$$(1 + \tau(t))\,\Phi'(e_s(t)) = p_e(s, e_s(t)) \cdot \left[\pi(n;t) - \pi(r;t) + q(t)\,(J_n(t+1) - J_r(t+1))\right], \tag{51}$$

with the equal-MC condition giving τ identical across firm types. We iterate to a fixed point; in the calibrated run the predicted-vs-realized residual converges below 10^{-6} and the equal-MC residual is below 10^{-4}. Tax revenue is rebated lump-sum and does not enter dividends.

No commitment, no front-loading. Figure 10 panel (a) plots three Q_n paths: CE (blue), static CEP (red dashed), and the Markov-perfect dynamic CEP (black). The Markov-perfect path is smooth: $Q_n^*(1) = 0.410$ (modestly below the CE's 0.424), declining gradually to about 0.373 in the terminal periods. There is no large initial overshoot of the kind that would only be optimal under credible long-horizon commitment. This is the natural consequence of dropping commitment: a planner who re-optimizes each period has no incentive to artificially elevate Q_n at $t = 1$, since doing so would commit her future self to undoing it.

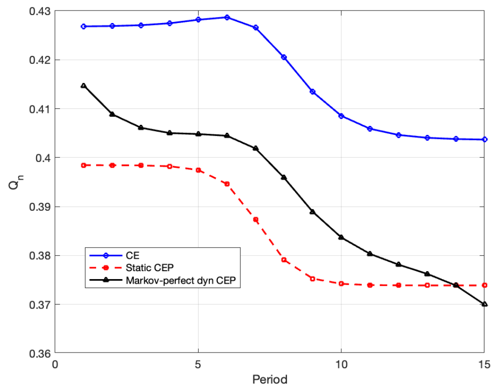
The Markov-perfect gap is flatter than the static gap. Panel (b) compares the over-innovation gap $Q_n^*/Q_n^{CE} - 1$ under the static and Markov-perfect dynamic CEP. The static CEP gap ranges from -6.8% to -10.2%, peaking during the transition. The Markov-perfect gap is flatter, ranging from approximately -3% to -7%, and does not exhibit the sharp widening at the transition peak. Intuitively, the Markov-perfect planner internalizes how today's Q_n shapes tomorrow's incentives, so the corrective wedge varies less sharply across the demographic shift.

Time-varying innovation tax. Panel (c) plots the implied innovation tax under both approaches. The static CEP tax stays in the 72–78% band, peaking at the start of the transition when ϕ_Y is highest. The Markov-perfect tax is more moderate and smoother, ranging roughly 60–63%, with a mild dip in the middle of the demographic shift before flattening out as the system approaches the planner's stationary steady state. The two approaches agree on the qualitative shape, but the static approach overstates the magnitude once dynamic continuation is taken into account.

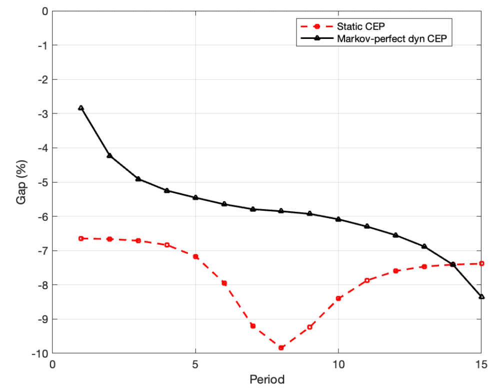
Welfare. Panel (d) shows that the Markov-perfect dynamic CEP achieves a small welfare gain over the CE (sum-of-flows gain of $+0.0030$), concentrated during the transition. The magnitude is modest in aggregate, consistent with the technology-composition externality being a second-order distortion: the level shift in Q_n matters more than the timing of corrections. The advantage of the Markov-perfect formulation is therefore not numerical magnitude but conceptual robustness—no commitment is required, the policy is a state-contingent tax that any future planner would maintain, and the headline tax of 60–63% is implementable as a rule indexed to (Q_n^-, ϕ_Y, ϕ_O).

6.5 Policy Implications

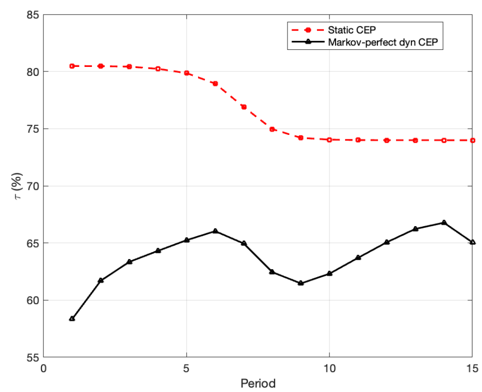
The technology-composition externality is inherently dynamic: its magnitude, welfare cost, and corrective instrument all vary with the economy's demographic state. Three policy lessons emerge from the quantitative analysis.



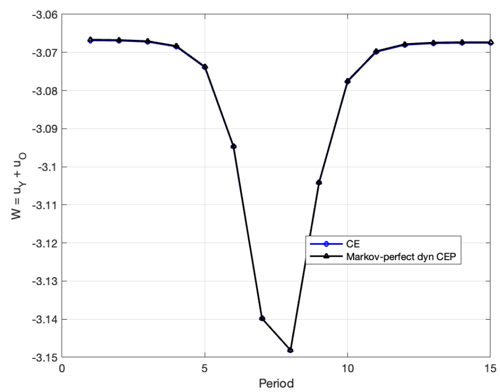
(a) Technology composition



(b) Over-innovation gap



(c) Implied innovation tax



(d) Welfare: CE vs. Dynamic CEP

Figure 10: Dynamic constrained-efficient policy along the transition (Markov-perfect).

Notes: Panel (a): Q_n paths under the CE (blue), static CEP (red dashed), and Markov-perfect dynamic CEP (black). The Markov-perfect path is smooth, with no front-loading: a planner who re-optimizes each period has no incentive to overshoot at $t = 1$ since the future self would not maintain the deviation. Panel (b): over-innovation gap $Q_n^*/Q_n^{CE} - 1$; the dynamic gap is flatter than the static gap. Panel (c): implied innovation tax; the static tax (72–78%) is overstated once dynamic continuation is taken into account, and the Markov-perfect tax (60–63%) is the implementable, time-consistent rule. Panel (d): aggregate welfare.

The case for intervention is strongest during demographic transitions. The externality peaks when demographics change fastest, not after the population has stabilized at its new age structure. The over-innovation wedge widens from -6.8% in the initial steady state to -10.2% at the transition peak (Figure 8, panel (c)), and the welfare gain from correction is concentrated in precisely these transition periods. This finding has a practical corollary: economies currently undergoing rapid aging—such as Japan, South Korea, and much of Southern and Eastern Europe—face the largest technology-composition distortions *today*, not in their demographic future.

Corrective policy should be time-varying and forward-looking. A fixed innovation tax, calibrated to either steady state, would be too weak during the transition and too strong afterward. The static CEP implies a transition-band tax of 72–78%, while the Markov-perfect dynamic CEP gives a more moderate 60–63% once the dynamic continuation is internalized. The Markov-perfect formulation is preferable as a policy rule: it is time-consistent (no commitment is required), it depends only on the realized state (Q_n^-, ϕ_Y, ϕ_O), and the implied tax varies smoothly with that state. In practice, such a rule could be implemented through R&D tax credits that are differentiated by the age profile of the target consumer base, or through procurement policies that steer public demand toward age-inclusive technologies.

Comparing these implied rates to actual R&D policy requires a margin distinction. Existing R&D tax credits in OECD economies—the U.S. R&D credit, the U.K. R&D Expenditure Credit, France's Crédit d'Impôt Recherche—typically offer modest subsidies of fifteen to thirty percent of qualifying expenditure. These instruments operate on the *level* margin: they target the overall quantity of R&D and are motivated by knowledge spillovers in the spirit of Romer (1990) and Aghion and Howitt (1992). Our corrective tax operates on a different margin: the *composition* of innovation between new- and retro-technology directions, conditional on the level. The two policies address distinct externalities and are not substitutes; they coexist. Concretely, our analysis implies a relative tilt between new- and retro-tech innovation of roughly $\tau_n - \tau_r \in [0.60, 0.83]$, which can be implemented as a differential within an existing R&D credit scheme rather than as a uniform tax that reverses its sign. The recommendation is therefore a relative re-weighting of innovation incentives, not a wholesale reversal of R&D policy.[8]

[8]Our model abstracts from the knowledge-spillover externality that motivates standard R&D subsidies, focusing exclusively on the technology-composition externality from age-driven demand heterogeneity. Integrating both externalities—a level wedge for spillovers and a composition wedge for demographic mismatch—in a single model is left for future work; the composition-margin lesson here is robust to the inclusion of a Romer-style spillover term in the planner's first-order conditions, since the two wedges enter additively. A direct cross-country exercise—plotting the new-vs-retro tilt of actual R&D-policy rates against the old-population share ϕ_O us-

A new form of dynamic inefficiency. The spike in the stochastic discount factor above unity during the transition signals dynamic inefficiency, but of a qualitatively different kind from the classical over-accumulation of capital (Diamond, 1965). The economy does not save too much—it innovates in the wrong *direction*. Unfunded transfers from young to old, the standard remedy for OLG dynamic inefficiency, would not correct this distortion: they redistribute wealth but leave the technology composition unchanged. What is needed instead is a technology policy that directly redirects the innovation mix. This distinction matters for policy design: countries that respond to aging solely through pension reform or intergenerational fiscal transfers will leave the technology-composition externality unaddressed.

Distributional considerations. The welfare cost of the externality is borne almost entirely by old households, while young households are nearly unaffected. This asymmetry suggests that the political economy of corrective policy may be favorable: the old, who stand to gain most from intervention, are a growing share of the electorate in aging societies. At the same time, the innovation tax reduces firm profits and hence the value of the asset portfolio that the young purchase—creating a potential intergenerational tension that any practical implementation must navigate.

7 Concluding remarks

This paper identifies a technology-composition externality in aging economies: the market over-innovates because no atomistic firm internalizes how collective innovation shifts the technology mix and relative prices against older consumers who bear adoption costs. The deeper structural source is that the technology choice itself carries no priced cost in the competitive equilibrium—both technologies share inputs at common factor prices, no specialized factor differentiates them, and no security or licensing market prices the aggregate composition Q_n. The corrective innovation tax is exactly this missing price. The externality is distinct from the classical dynamic inefficiency of over-accumulation—the economy produces the

ing the OECD R&D Tax Incentives Database—would document the composition-margin mismatch quantitatively. We leave this empirical exercise to a companion paper.

wrong *kind* of technology, not the wrong quantity of capital—and cannot be corrected by intergenerational transfers alone.

Four quantitative findings stand out. First, the competitive equilibrium produces 7%–10% more new-technology firms than the static-CEP allocation, with the gap widening during the demographic transition; under the Markov-perfect dynamic CEP the gap stays in a flatter 3%–7% band. Second, the welfare cost falls almost entirely on old households, who inherit a technology mix shaped by yesterday's larger young cohort, amplifying intergenerational inequality precisely when aging is fastest. Third, the optimal corrective policy is a state-contingent innovation tax of 60%–63% along the calibrated transition, with the static-CEP analog higher at 72%–78%. Fourth, and the empirical core of the paper, a sensitivity analysis across demographic states shows that the corrective tax stays in a narrow 69%–83% band as the population size rises from 0.8 to 1.5, with the absolute externality wedge between social and private marginal cost approximately constant: the technology-composition distortion is structurally large and robust across demographic regimes, yet no actual economy implements a tax in this range. The same analysis shows that aging depresses innovation effort by roughly a third and shrinks the new-technology firm share, while the over-innovation gap relative to the planner widens—suggesting that the case for corrective policy is strongest precisely in the aging economies where innovation is already declining.

The results carry a broader message. As populations age across the developed world, the mismatch between market innovation incentives and the needs of older consumers is likely to grow. Policies that address aging solely through pension reform or fiscal transfers will leave the technology-composition distortion unaddressed. The case for technology policy—steering the *direction* of innovation, not just its level—becomes increasingly urgent as demographic transitions accelerate.

Several extensions merit future work. First, richer lifecycle heterogeneity within the old cohort—varying adoption costs by health, education, or cohort—would allow the model to capture within-generation distributional effects and speak to the heterogeneous experience of aging across socioeconomic groups. Second, endogenous quality improvement within technology types would distinguish between innovation that creates genuinely new products and incremental upgrades to existing ones, potentially moderating the externality if quality improvements benefit all consumers. Third, exploring the interaction between innovation policy and intergenerational transfers would clarify whether the two instruments are complements or substitutes. Finally, extending the model to an open-economy setting would allow for cross-country technology spillovers, where the technology mix in one country—shaped by its demographic structure—affects the product variety available to consumers in another.

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